Conditional generation of visible language

a Training

60 visible glyph images
Frozen causal reader

→

Continuous visual history
384 state + 1024 image field

History conditions every residual scale

Four noisy future cells
Independent or shared noise levels

→

Conditional strip denoiser
FiLM and cell attention

→

Velocity and visual losses
5.90M trainable parameters

b Inference

Noise and visible history

→

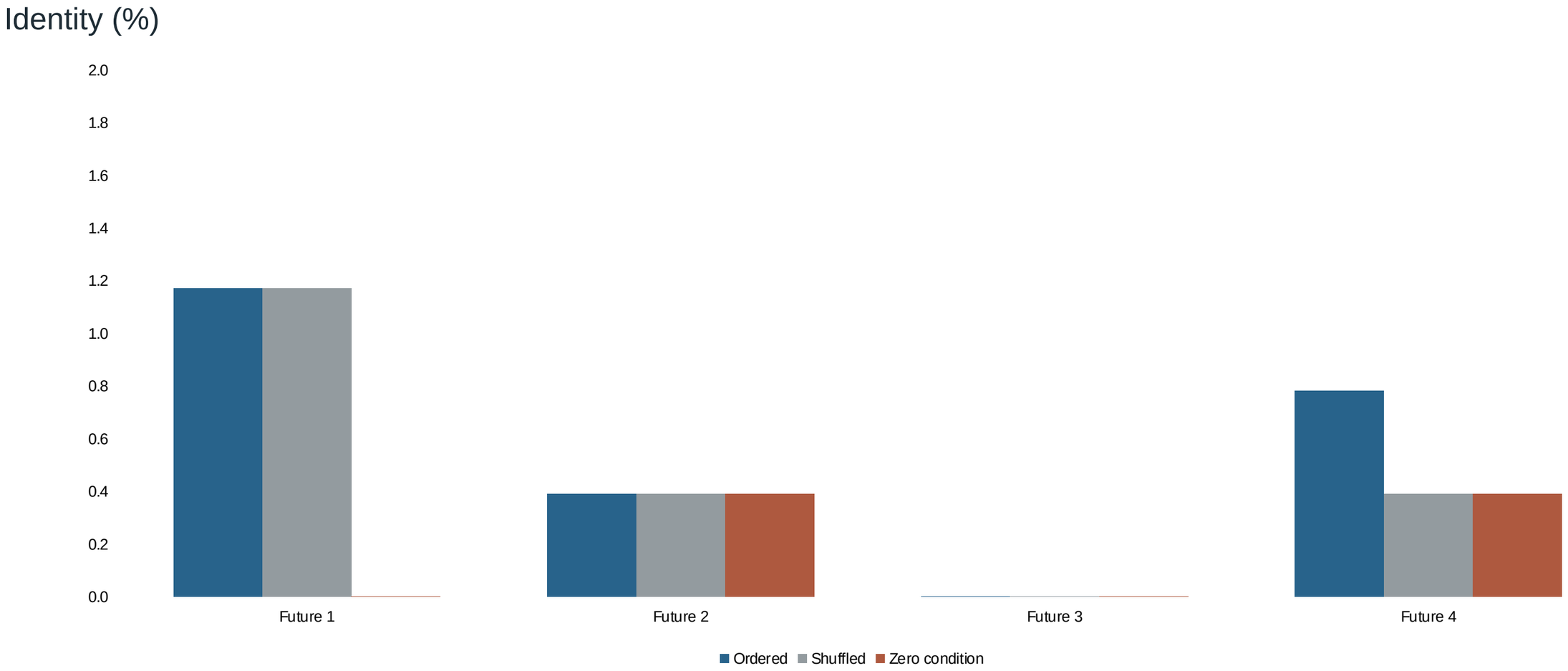
32-level reverse process

→

Four output images
Append pixels to next context

Source: V53 frozen protocol, implementation and locked audit. Schematic objects are editable.

Direct block generation under paired visual histories



256 prefixes, one primary sample each, four output cells
Ordered block identity 0.586%, shuffled 0.488%, zero condition 0.195%

Source: locked V53 development report. Native chart includes its editable evidence workbook.

Unselected output images expose the language error



Fixed first eight audit rows. Pixels are model output, with nearest-neighbour enlargement. No glyph replacement.

Exact raster likelihood preserves a visible copy path

64 writing images
Frozen 16.28M reader

→

Continuous visual state
Hidden384 + image field1024

→

Fixed inverse field
Aligned signed pixel plan

Training: exact Bernoulli image NLL

Earlier target rows
Four-layer row transformer

→

Earlier pixels within row
Strictly masked MADE

→

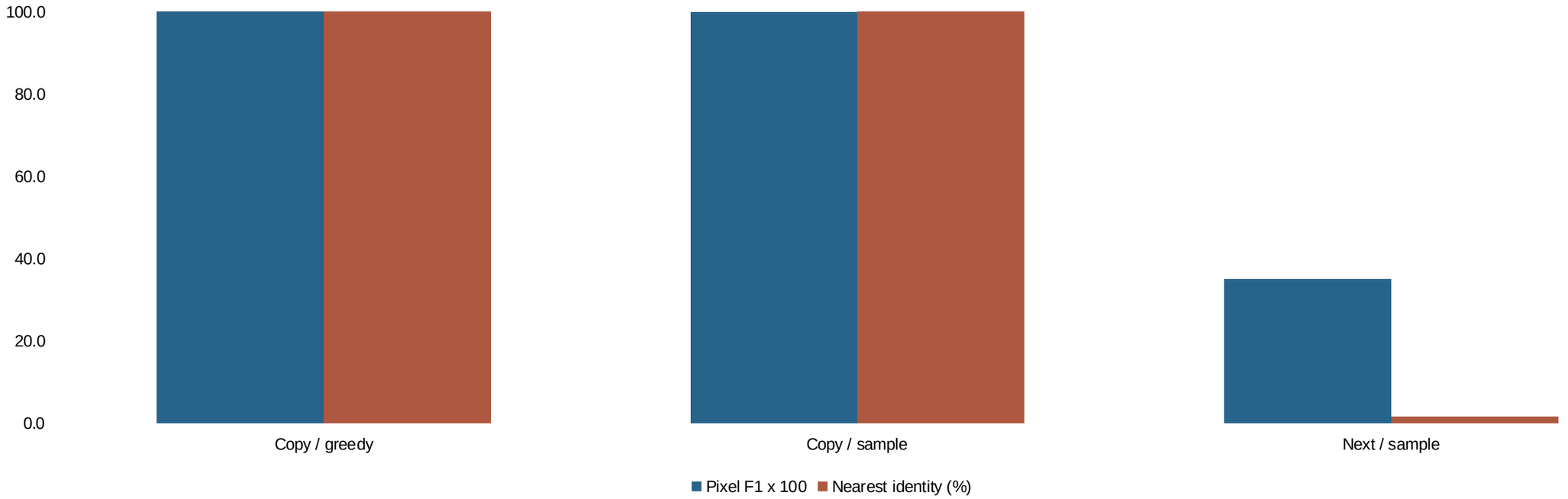
Aligned logit + residual
Normalized next-image law

25% supplied-image motor exercises; 75% natural predicted conditions. No glyph inventory.

Inference: sample 1,024 causal pixels; append the visible raster to history

2.43M new density parameters; 18.71M total. Supplied-image copying is architectural, not language evidence.

Copying succeeds; autonomous language remains unresolved



Exact-image copying: 256/256 greedy. Autonomous target identity: 4/256 ordered, 3/256 shuffled.

Target NLL: 15.219 ordered; 15.254 shuffled nats/image
Paired advantage 0.0355 nat; source-record bootstrap 95% CI [-0.1014, 0.1693]

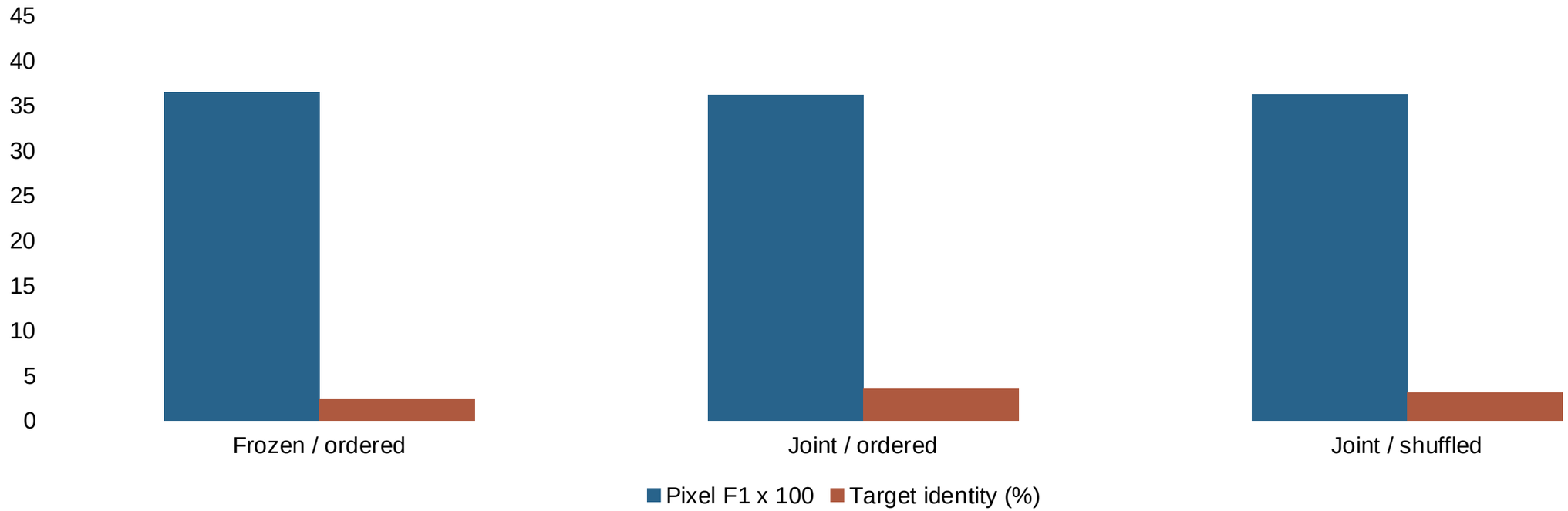
V54 locked development audit. Native chart and embedded workbook. Supplied-image controls reveal the answer.

Actual rasters: exact motor copying and failed continuation



First eight fixed audit rows; nearest-neighbour enlargement. No OCR, image replacement or selected examples.

Matched joint learning: better likelihood, incomplete generation

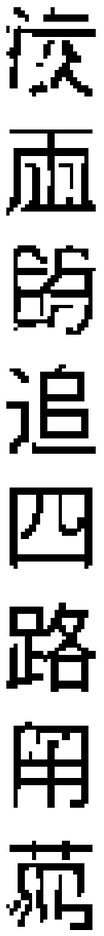
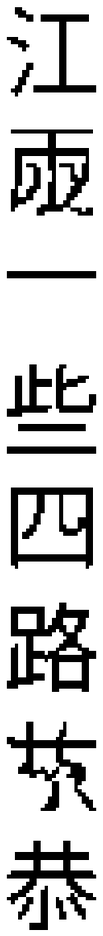
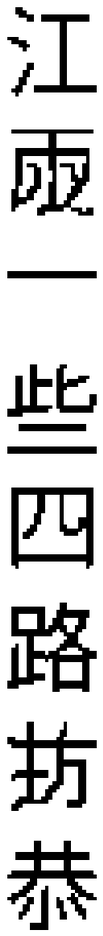


Exact target NLL: joint 7.026; matched frozen 7.579 nats/image
Paired gain 0.553; 95% source-record interval [0.450, 0.665]

Both arms retain exact supplied-image copying. Joint model passes 6/8 gates; generation remains unqualified.

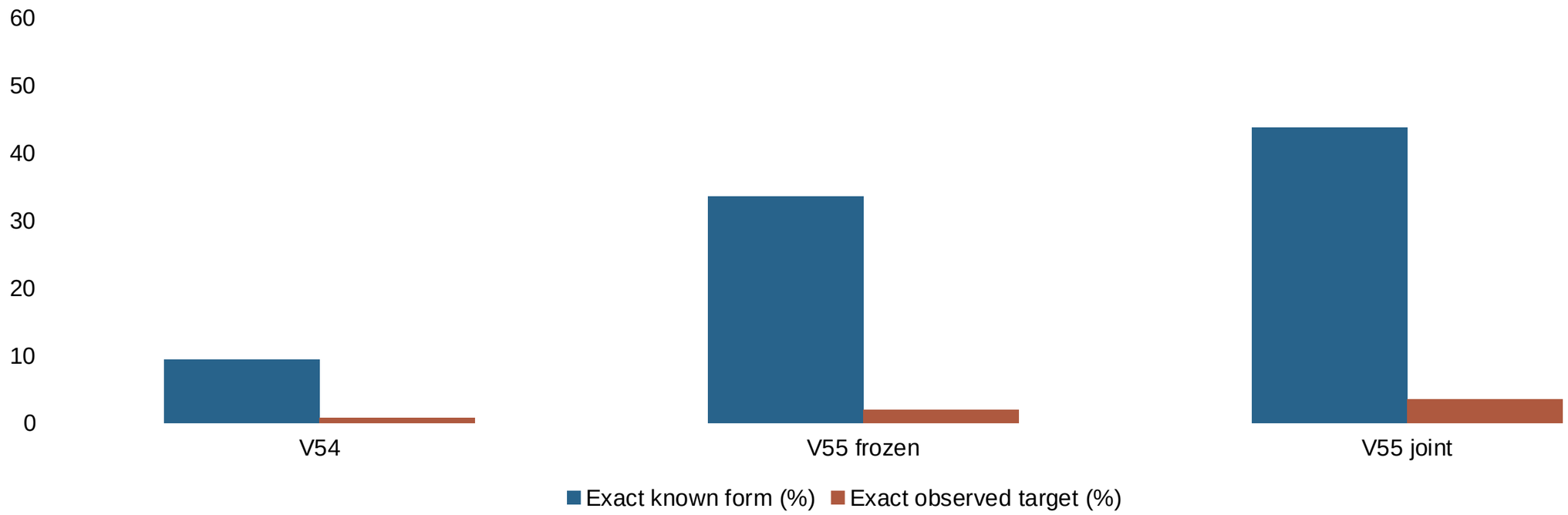
V55 locked matched audit; 512 NLL targets, 256 single-sample prefixes. Native chart includes its editable workbook.

Raw matched output: exact copying and remaining language errors

Target	Frozen sample	Joint sample	Joint shuffled	Joint greedy	Supplied copy
					

First eight fixed audit rows, no selection. Pixels are actual output; no OCR, evaluator-bank replacement or external inference.

Clear glyphs and correct language are separate tests

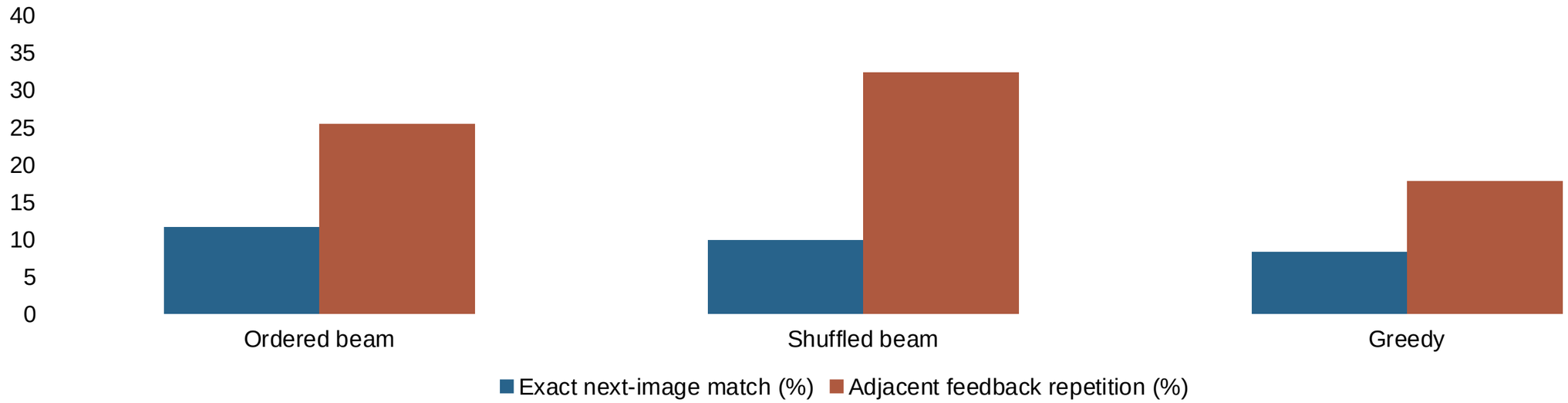


Exact known-form gain, joint minus frozen: 10.156 percentage points
95% source-record interval [4.219, 16.240]; 256 saved first samples

The fixed 9,932-image bank is used only for evaluation. No image is repaired or replaced.

Post-result diagnostic; all original qualification decisions remain unchanged. Native chart includes its editable workbook.

Raw-pixel beam search: better immediate choice, repetitive output

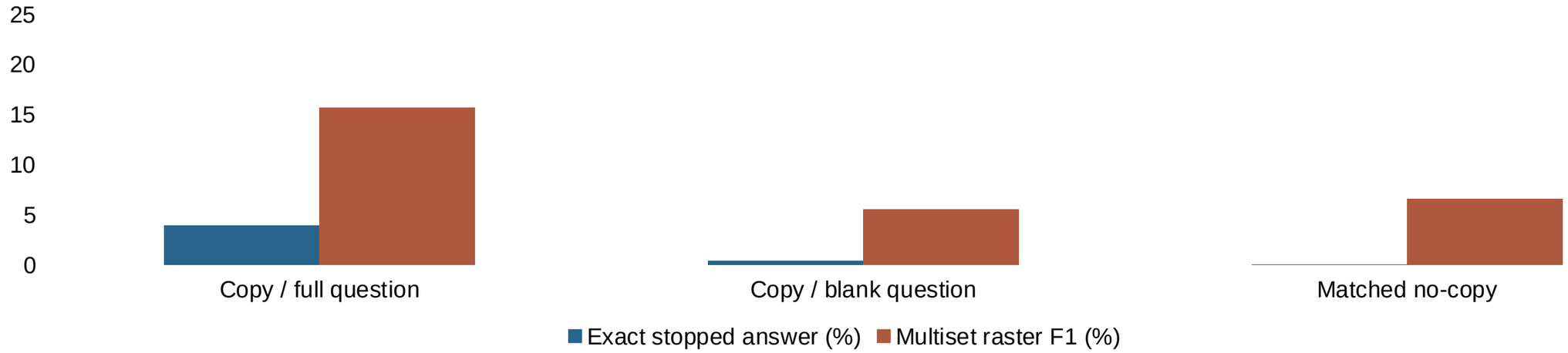


2,048 prefixes: beam gain over greedy 3.369 points [2.351, 4.412]; over shuffled 1.758 [0.966, 2.552]
Observed-stream repetition: 0.112%; beam: 25.446%. Rollout gate fails; decoder rejected.

Observed future	Ordered beam	Shuffled beam	Original greedy
不尬的就跑回来了	一个，不是一个，	一个，不不不不不	，不不一，不不不
的女婿。丈人教我	，不是一人，不是	，不不不不不不不	，不不不不不不不

Unselected first two eight-step audit streams. Unchanged V55 weights; no vocabulary or repair. Both controls use highest FP32 precision.

Natural Chinese questions: some binding, inaccurate complete answers



Question-dependent F1 gain: 10.165 points [7.043, 13.371]
Exact answers: 10/256; both questions correct: 1/393 pairs. Four of seven gates pass; model rejected.

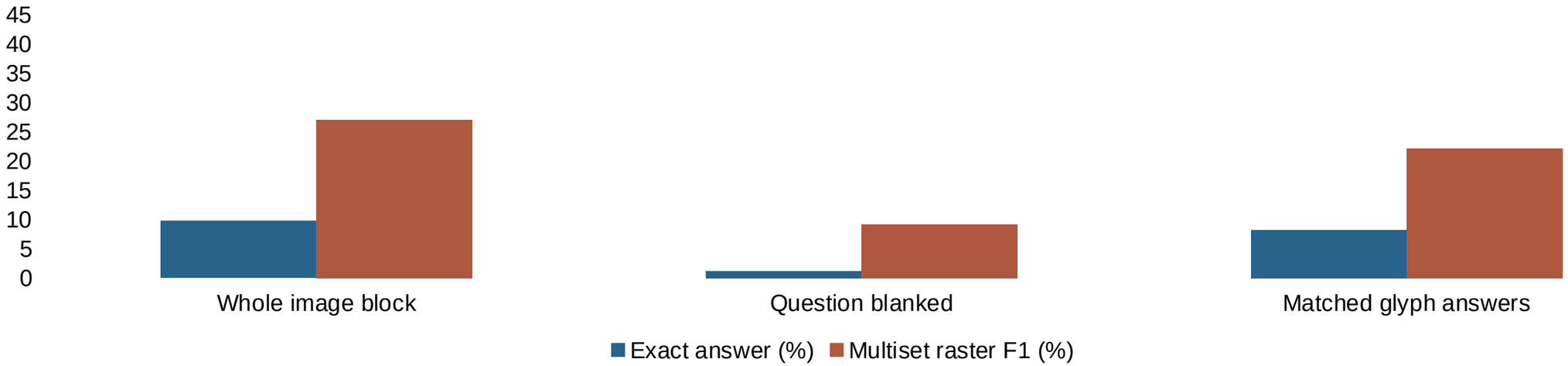
Q1 １９４０年８月世宗大学更名成什么？

Q2 邵伯温是在什么时候去世？

Reference 1	Copy	Blank question	No-copy
首尔女子师范学院 高宗绍兴四年（１１３４年）	１９４０年５月，Ｙ月世宗大学” 邵溥、邵博、邵传后》	节居绍圣、利州路转运作）	１９３０年 １３３０年

First two fixed audit rows; actual output images. White means zero emitted cells. All primary copy-arm answer cells use observed prompt images.

Whole observed image answers: better F1, unresolved exact-match gain



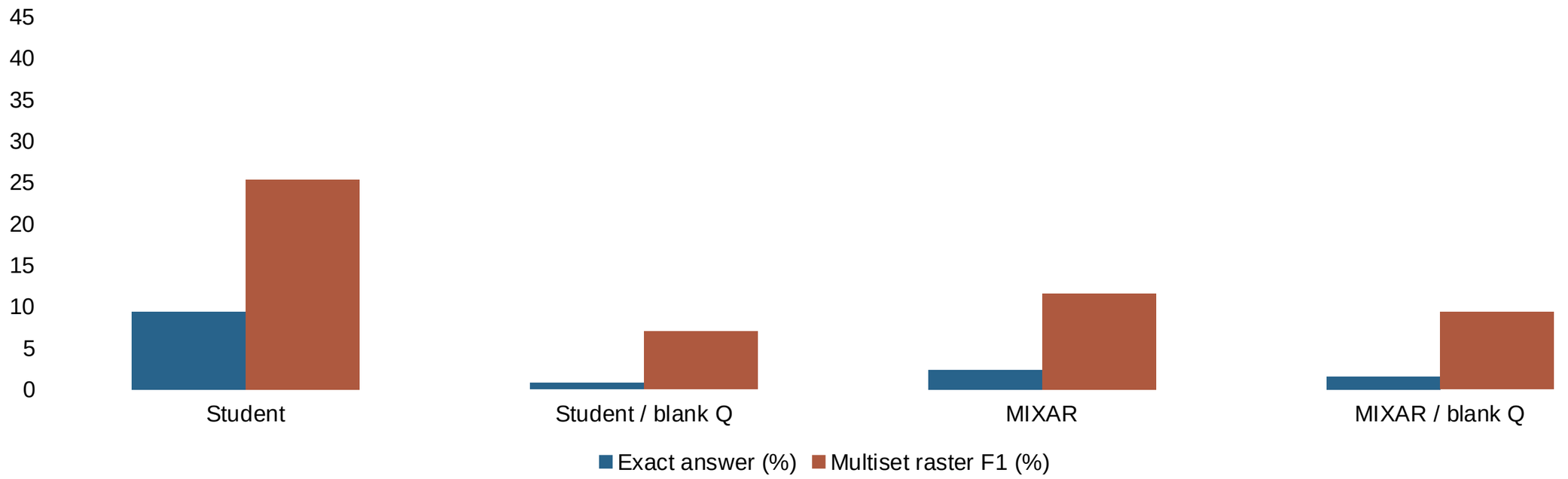
25/256 exact answers; matched glyph control 21/256. F1 gain 4.893 points [0.469, 9.047].
Only 4/393 paragraph pairs have both answers correct. Six of seven gates pass; reading rejected.

Prompt pixels → Causal reader → Start / end law → Source image answer

	Retained reference	Actual block answer
Audit row 1	首尔女子师范学院	“世宗学院”
Audit row 2	高宗绍兴四年（1134年）	卒于高宗绍兴四年（1134年）

First two fixed audit rows. Copying preserves selected pixels; it does not certify the meaning of the selected text.

Frozen visual foundation probe: larger pretrained features do not help here

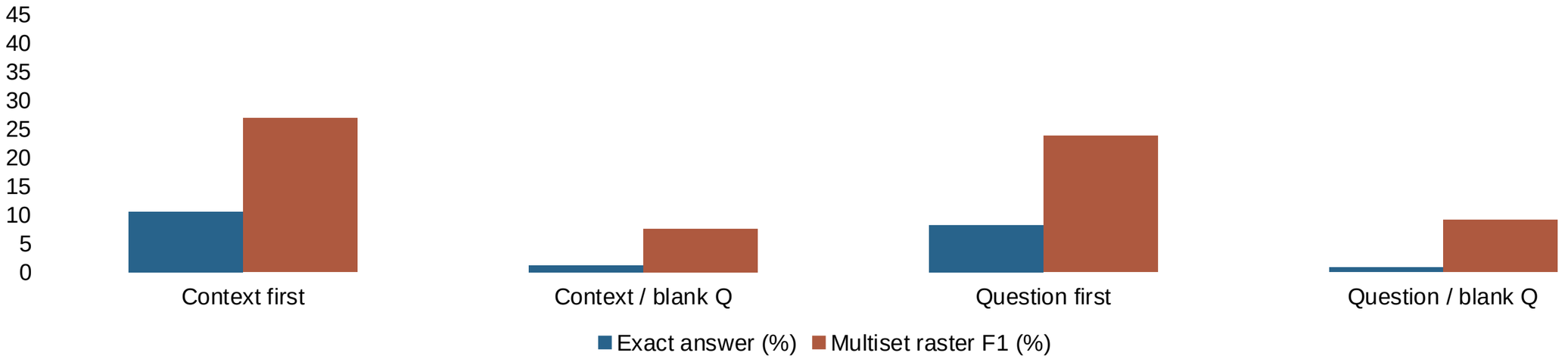


Identical 789,520-parameter heads; 4,000 matched updates each.
Student 24/256 exact; MIXAR 6/256. MIXAR-minus-student F1 −13.752 points [−18.247, −9.094].

MIXAR has different pretraining and native rendering; this probe uses our Noto CJK grid.
External four-H200 pretraining is not one-4090 work. Frozen readout comparison rejected; not a general encoder ranking.

Continuous prompt-image features only. Shared cached release weights; no external inference call or answer annotation enters feature preparation.

Does reading the question first improve evidence selection?



Context first

Passage images X → Question images Q → Answer

Question first

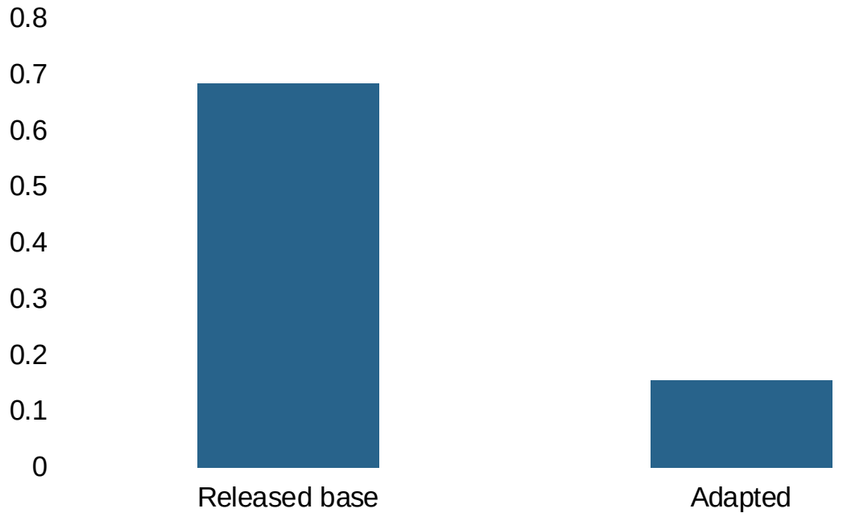
Question images Q → Passage images X → Answer

Question-first minus context-first F1: -3.143 points [-6.544, 0.446].
4/8 registered gates pass; decision: reject_question_first_intervention.

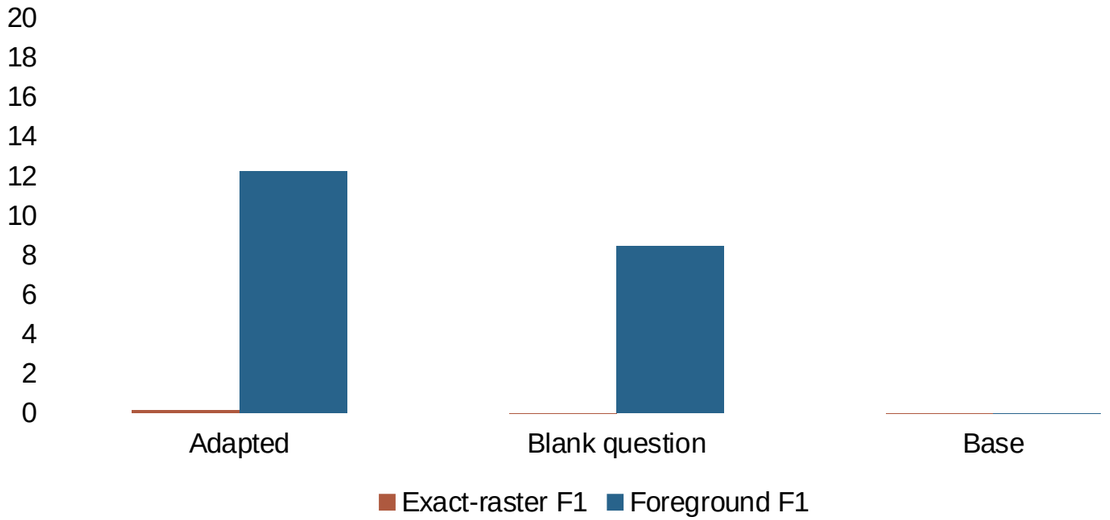
Same checkpoint, data, parameters and training budget. Source pixels only; reused development, not a sealed final test or general language qualification.

Native free writing: improved pixel fit, failed language generation

Held-out answer-pixel NLL (nats)



Generated image scores (%)



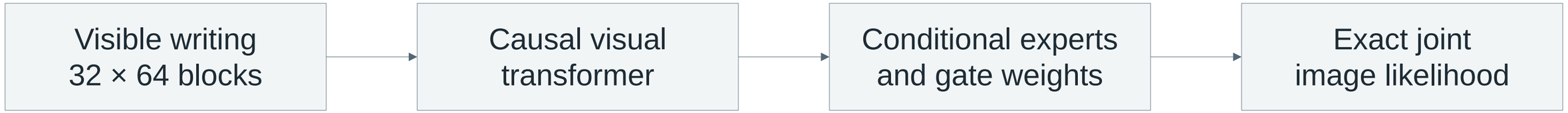
0/256 exact answers and 0/393 pairs both correct. Five of ten registered gates pass.
477.15M externally pretrained parameters, fully adapted for 1,000 updates on GPU 0.

	Retained reference	Actual free-pixel answer
Audit row 1	首尔女子师范学院	
Audit row 2	高宗绍兴四年（１１３４年）	

Training: 1,501.1 s and 9.912 GiB. Offline native-PNG replay passes. External pretraining and reused development disclosed.

Conditional whole-image density

Training: prompt and observed answer images



$$p(y \mid h) = \Sigma k\ \pi k(h)\ \Pi i\ \text{Bernoulli}(yi\ ;\ \text{sigmoid}(aki(h)))$$

One shared expert choice couples all pixels in a bitmap. Each expert is a neural function of context.
K = 1: 477.15M parameters. K = 8: 490.00M. Same examples and update budget; different head size.

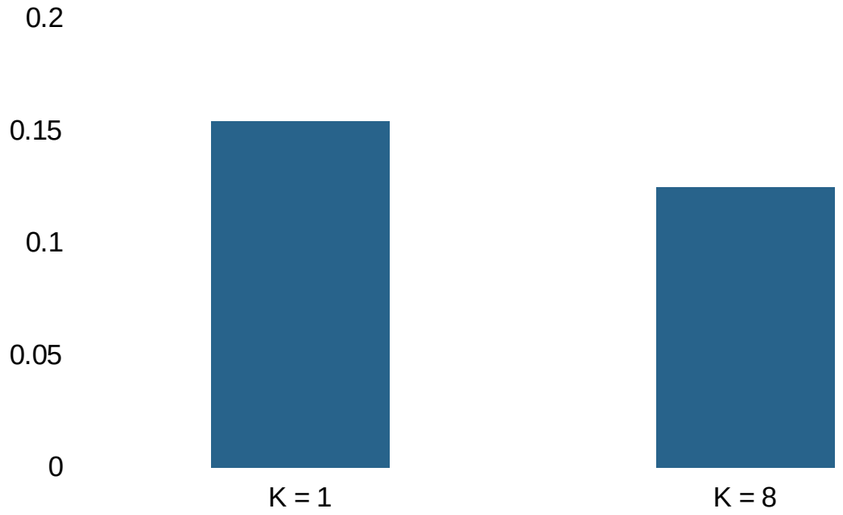
Inference: image prompt → candidate bitmaps → score under full density → physical pixels

Feed generated pixels back into the visual context; stop at an all-white image block.
Component-mode search is approximate MAP. No character bank, OCR, Unicode IDs or external LM in the actor.

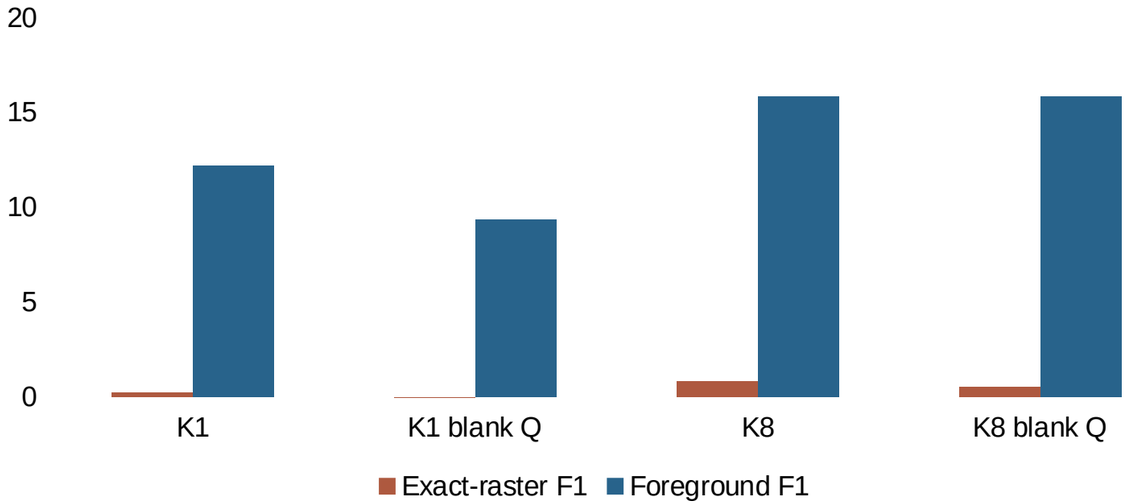
Externally pretrained MIXAR foundation. The mixture law is a tested hypothesis, not a guarantee of readable language.

Conditional density: held-out fit and generated writing

Held-out answer-pixel NLL (nats)



Generated image scores (%)



Exact answers K1 / K8: 0 / 0 of 256. Both-correct pairs: 0 / 0 of 393.
Registered decision: reject density intervention. General image-language quality remains unqualified.

	Retained reference	K = 1	K = 8
Audit row 1	首尔女子师范学院	[illegible]	1 9 4 0 年
Audit row 2	高宗绍兴四年（1 1 3 4 年）	[illegible]	1 9 [illegible]

First two fixed primary answers, unrepaired; white display padding. One seed, reused development, external pretraining disclosed.

Question-conditioned image-language training

a Observed training images: two questions about one paragraph

Question region Q0

哪个人是世纪越南反唐起义军的军事首领？

Recorded answer Y0 冯兴

Question region Q1

冯兴与他的弟弟冯骇接受谁的建议，率众起义？

Recorded answer Y1 杜英翰

Each full prompt Xi also contains the same paragraph image. Appended white termination blocks are not drawn.

b Four causal likelihoods under the same parameters

(X0, Y0) (X0, Y1)
(X1, Y0) (X1, Y1)

→

Shared image generator
490M parameters, 8 experts

→

	Y0	Y1
X0	e_{00}	e_{01}
X1	e_{10}	e_{11}

eij = mean negative log likelihood of the complete answer suffix, including termination

c Positive image likelihood with paired question assignment

$$D = e_{01} + e_{10} - e_{00} - e_{11}$$

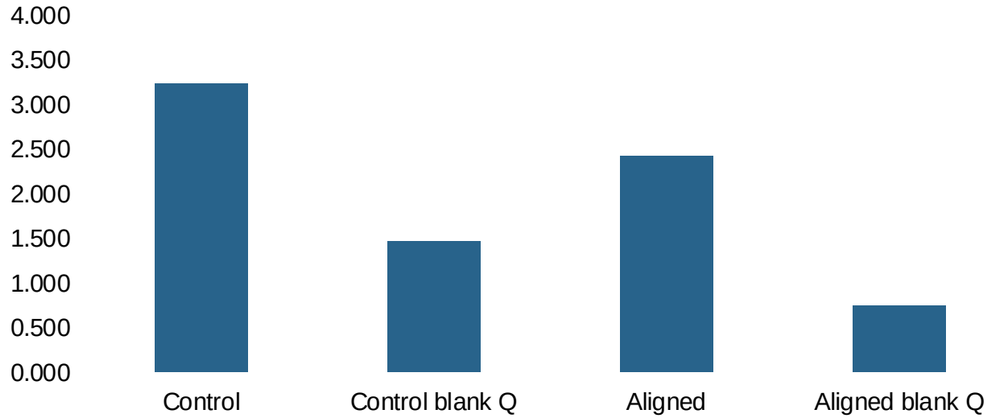
λ = 0: control; λ = 1: assignment

$$L = (e_{00} + e_{11}) / 2 + \lambda \cdot 0.01 \cdot \mathrm{softplus}(-D / 0.01)$$

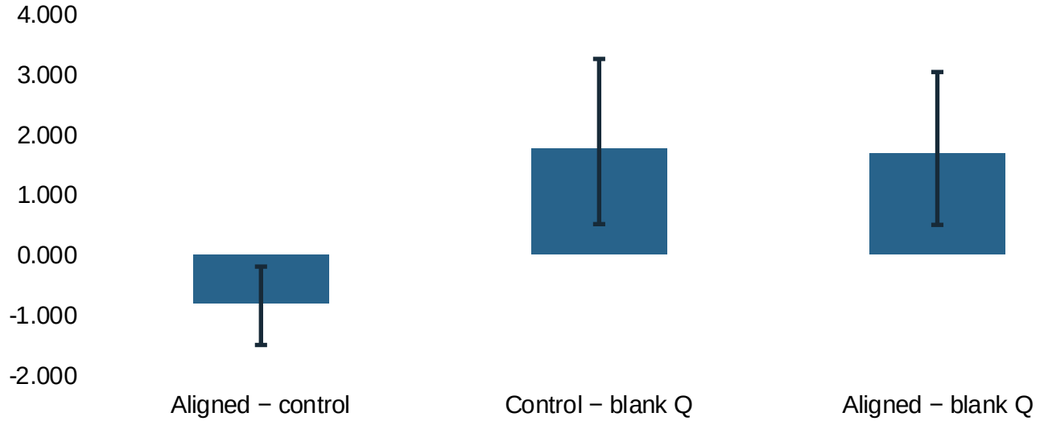
Observed targets, not model outputs. Both arms start from the same V62 EMA; autonomous answers are evaluated separately.

Question assignment: completed free-writing test

Strict-raster F1 (%)



Paired differences with 95% intervals (pp)



Exact answers: control 2/256, aligned 1/256. Both-correct pairs: 0/393 in each arm.

First two fixed primary answers

1: Reference 首尔女子师范学院

1: Control



1: Aligned



2: Reference 高宗绍兴四年（1134年）

2: Control



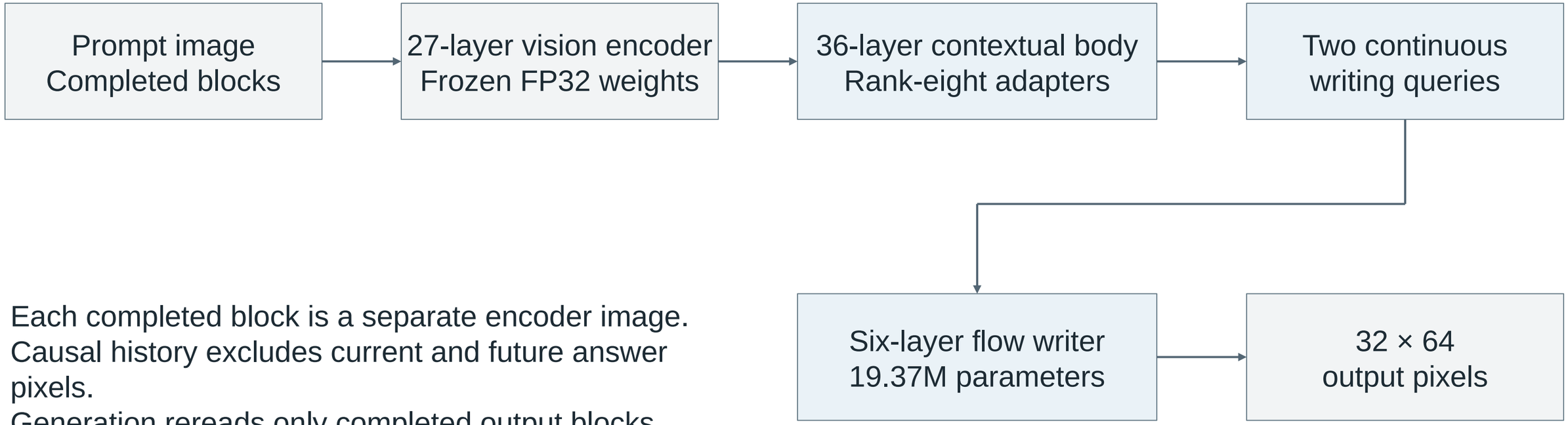
2: Aligned



Unrepaired pixels and white padding. One seed, reused development and external pretraining. Both reading gates fail.

Complete continuous image actor

7.548B total parameters 25.58M trainable 17.181 GiB peak allocated



Each completed block is a separate encoder image.
Causal history excludes current and future answer pixels.
Generation rereads only completed output blocks.

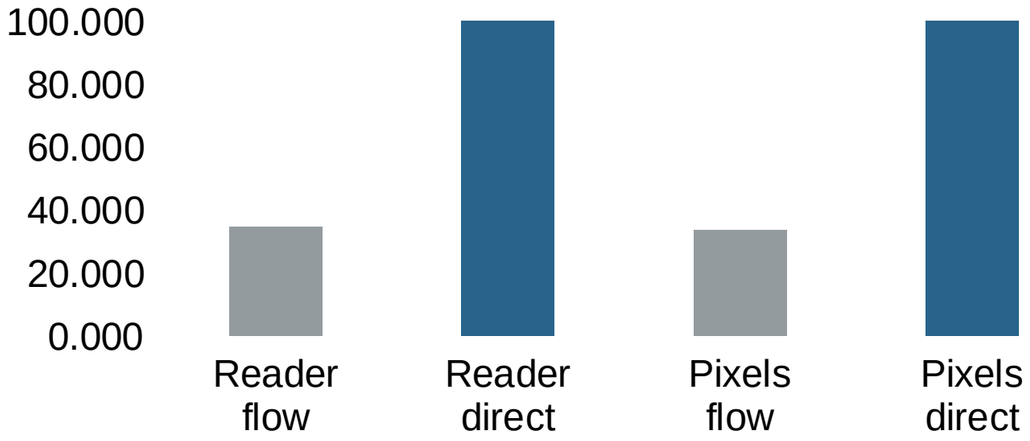
$$Yt = (1 - t)\,U + t\,Y \qquad U \sim N(0, I) \qquad Lflow = E\,||v(Yt,t,h) - (Y - U)||^2$$

Frozen contextual weights use BF16 storage with FP32 computation. A separate classifier predicts stopping.

Execution checks pass. Writing and language quality remain unqualified.

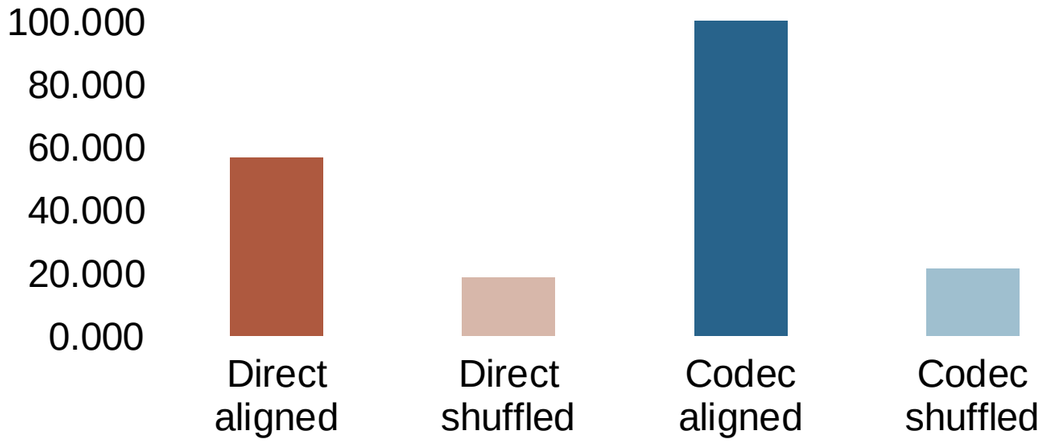
Writing fidelity depends on the output representation

a Training F1 (%) | n=64



Exact blocks: 0/64, 63/64, 0/64, 64/64

b Development F1 (%) | n=128



Exact aligned: direct 3/128; fixed codec 126/128

c First two fitting rows

	Reference	Reader flow	Reader direct	Pixels flow	Pixels direct
1					
2					

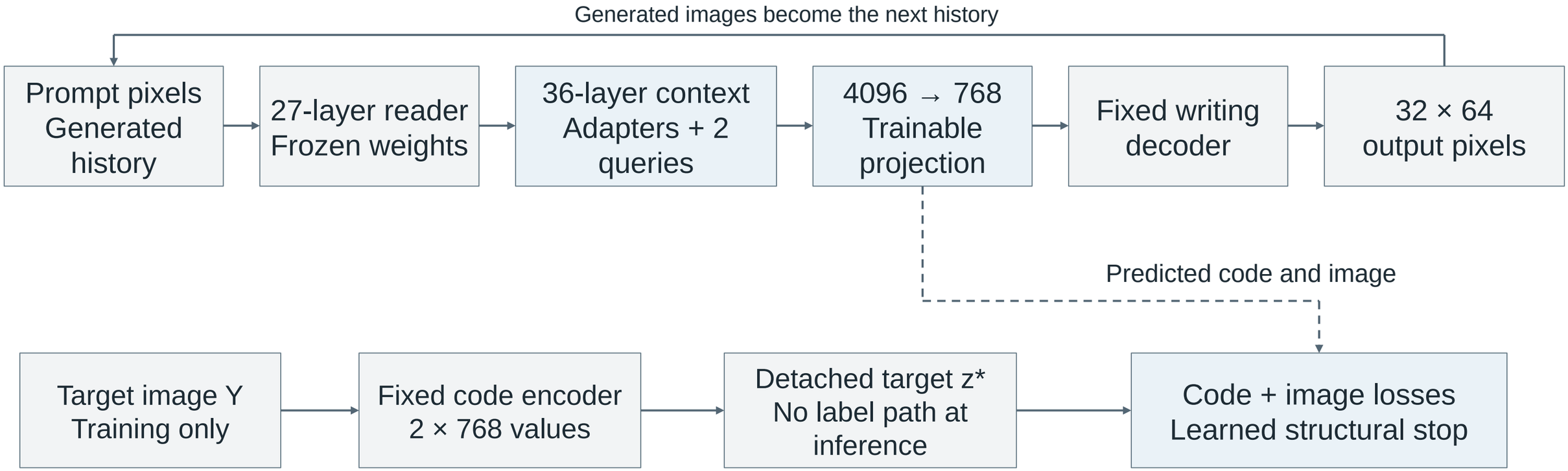
d First two development rows

	Reference	Direct	Codec
1			
2			

Fitting uses training images. Development compares supplied representations with unequal prior training; neither is question answering.

Reading and writing use separate continuous spaces

7.539B total parameters 9.352M trainable 16.953 GiB peak allocated



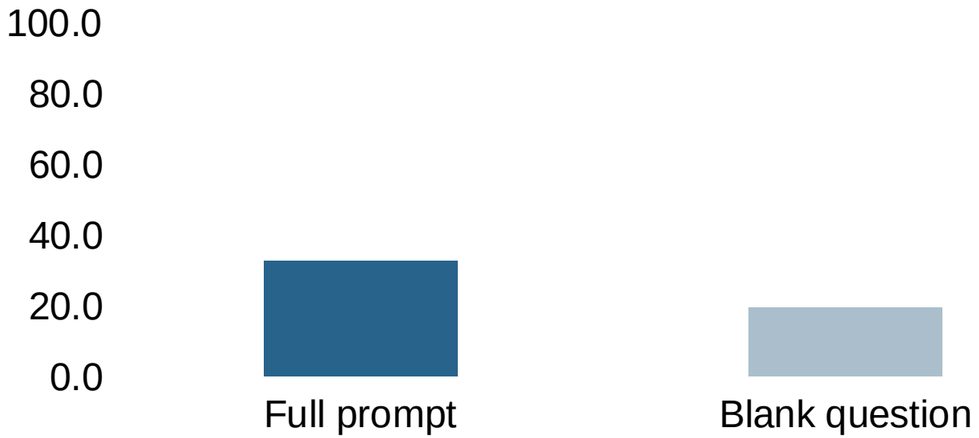
$$L = MSE(N(\hat{z}), N(z^*)) + MSE(D(\hat{z}), Y) + Lstop$$

Frozen decoder weights still pass gradients into the predicted codes and contextual adapters.

All 24 complete-model execution checks pass. Two-update output is fragmented; autonomous language remains unqualified.

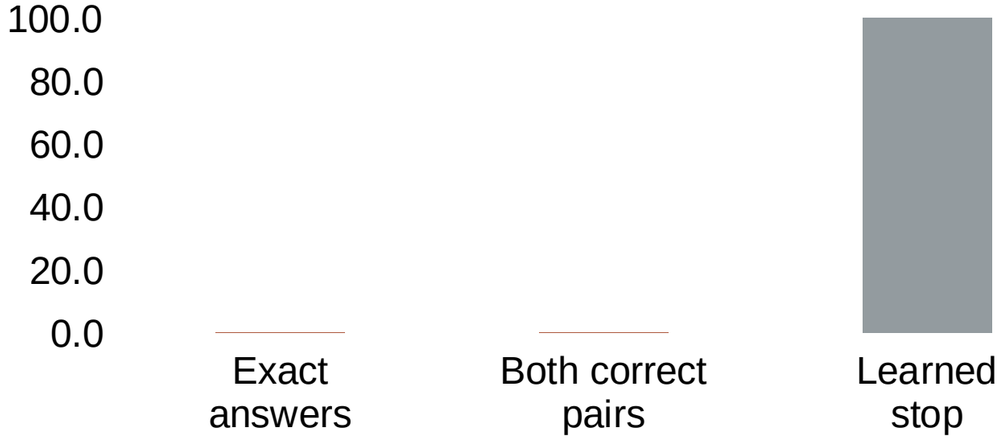
Question effects persist when accurate writing fails

a Chinese foreground F1 (%) | n=128



Paired gain: 13.16 points [10.93, 15.73]

b Exact answers and learned stopping (%)



0/128 exact answers; 0/64 complete pairs; 128/128 stops

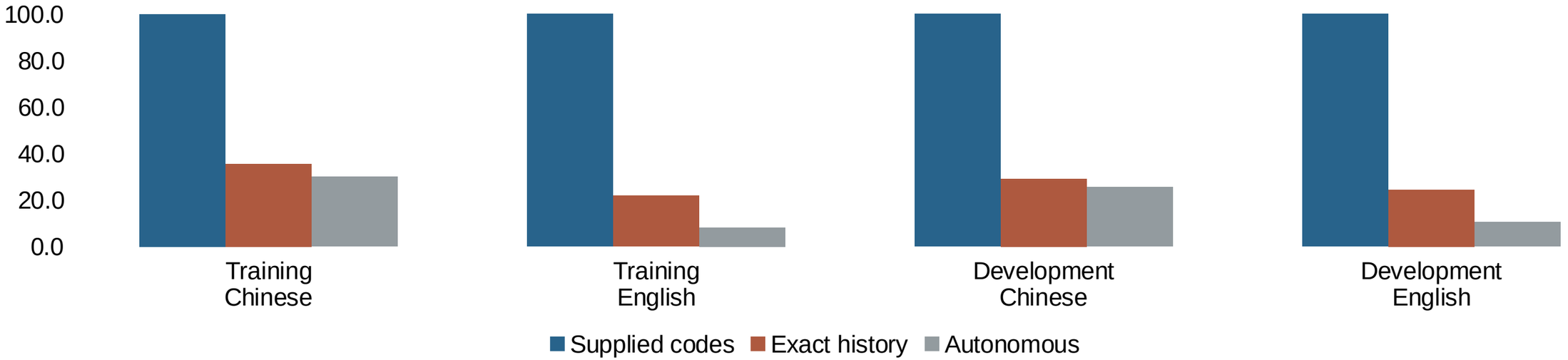
c First two fixed Chinese cases | complete, unmodified streams

	Reference	Full prompt	Blank question
1	玛丽莲·曼森	[illegible]	[illegible]
2	批评他的人	[illegible]	[illegible]

All 384 generations and original gates are verified. Question dependence and stopping pass; answer quality fails.

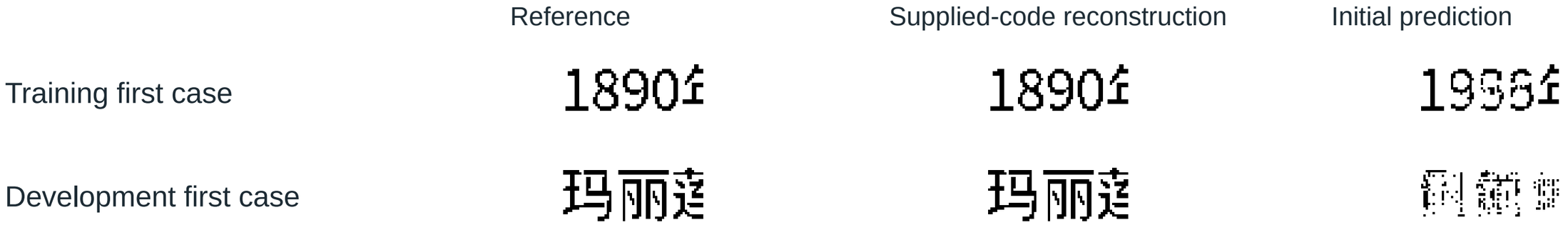
Prediction fails before generated-history drift

a Foreground F1 (%) | eight fixed cases per group



Exact history supplies previous images and answer length; it is an assisted diagnostic.

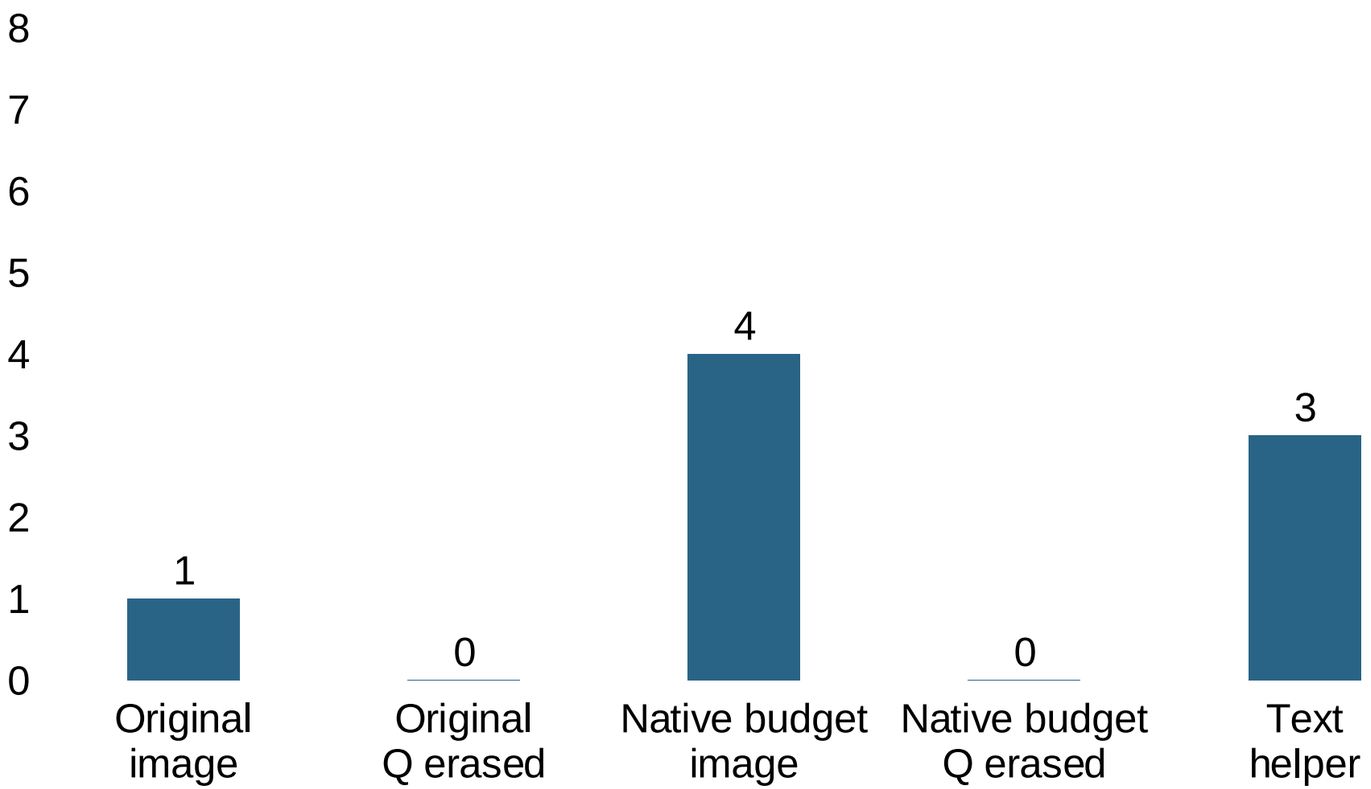
b First 32-by-64 block, before generated history



No first predicted block is exact. All 32 initial predictions match the autonomous path; no diagnostic weights update.

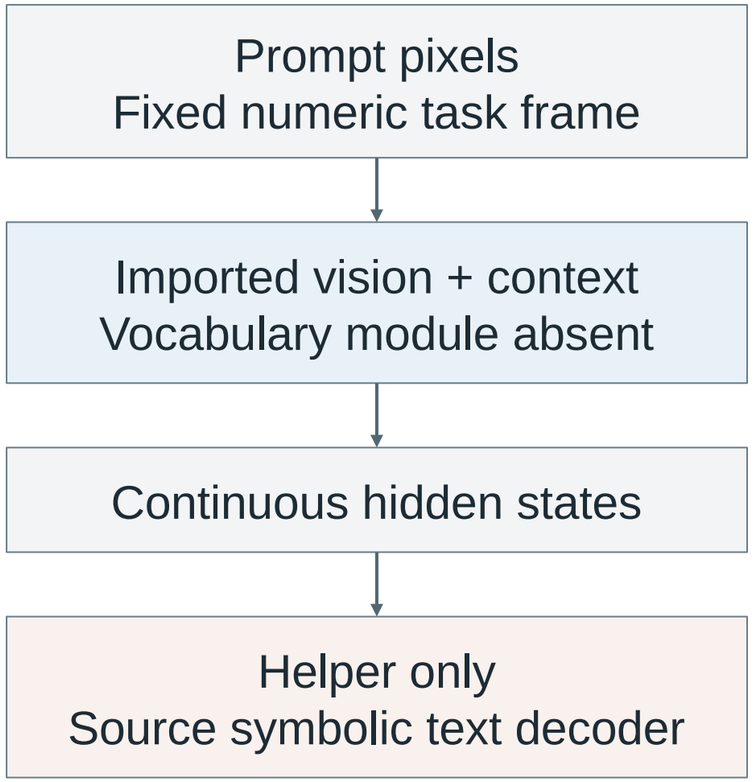
Numeric image controls retain bounded reading

a Exact Chinese answers | the same eight reused questions



Q erased: only question ink is removed, with wrap coordinates retained.

b Diagnostic boundary

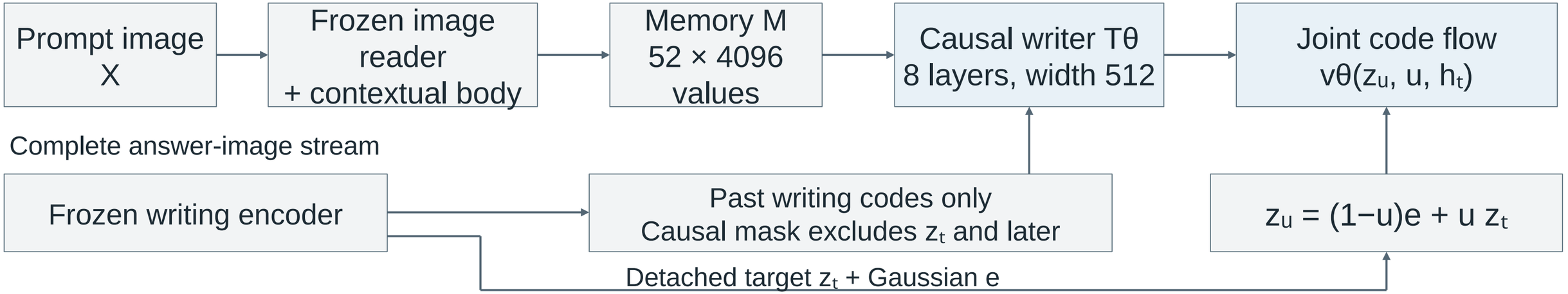


Text decoding is excluded from the image actor.
This check generates no image answers.

All 32 input images and 742 retained decoding choices verify. The complete neural forward pass is not independently replayed.

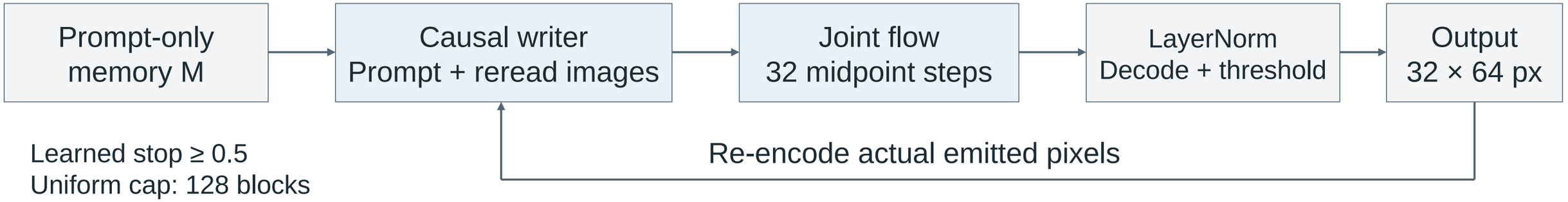
Causal image writing with frozen visual semantics

a Training | complete answer images supervise continuous writing



$$L = \text{mean } ||v\theta - (z_t - e)||^2 + \text{balanced stop loss}$$

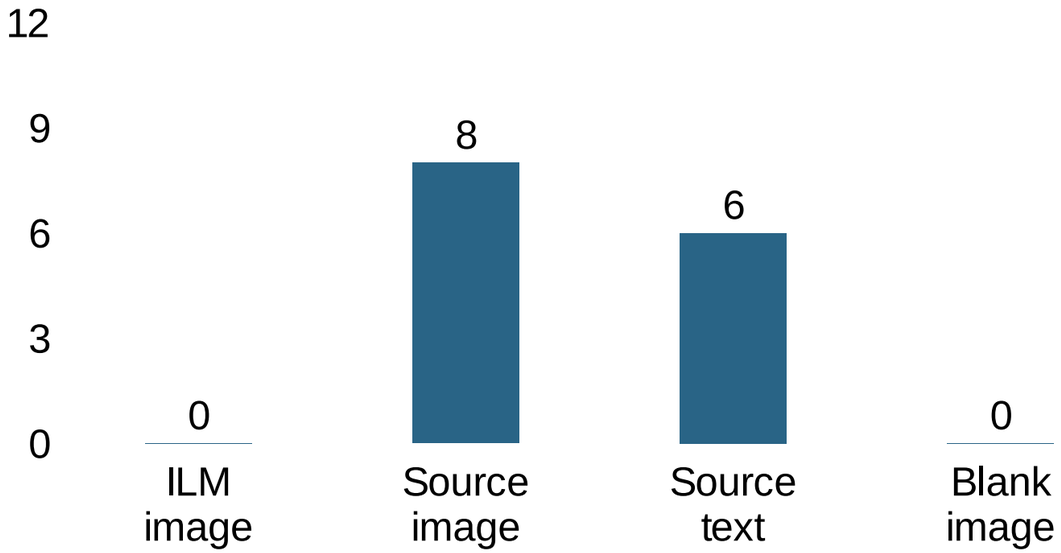
b Inference | every generated block becomes physical visual history



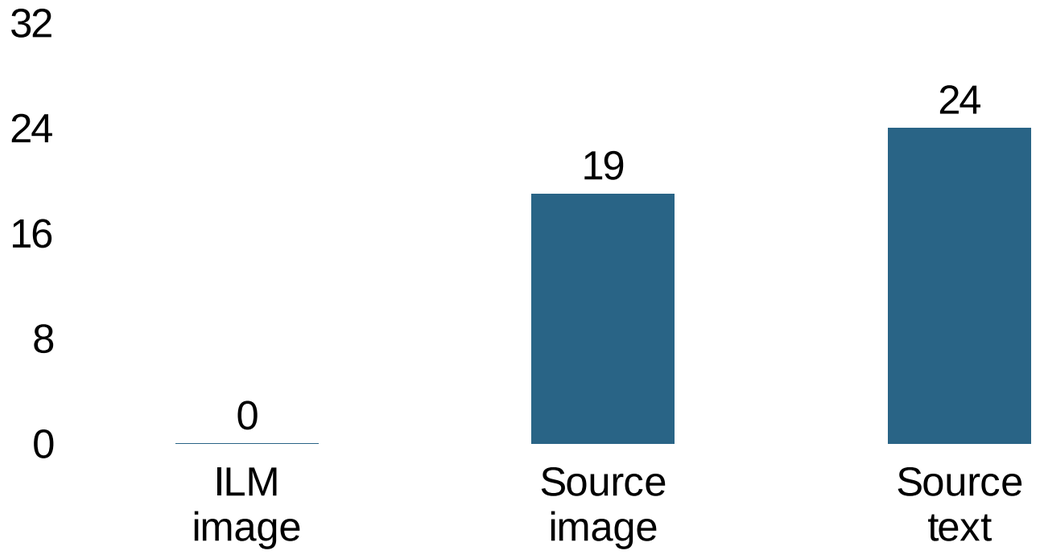
38.65M trainable writer. External reader pretraining retained. Completed Chinese audit: 0/128 exact; 29.386% foreground F1.

Reader content before image writing

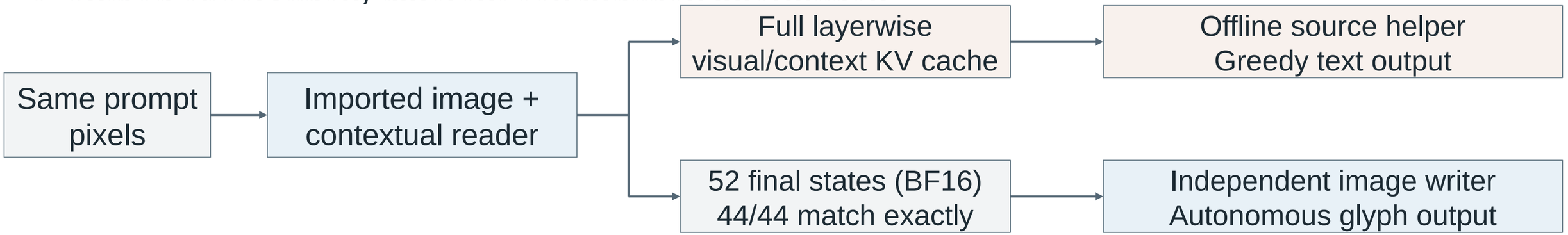
a Exact copied answers / 12 new requests



b Exact answers / 32 fixed training questions



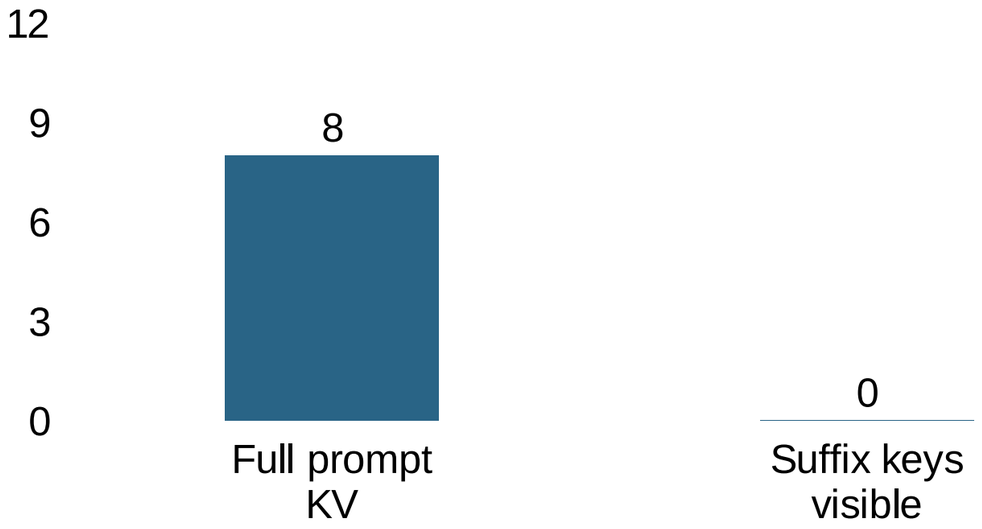
c Shared observation; different continuation information



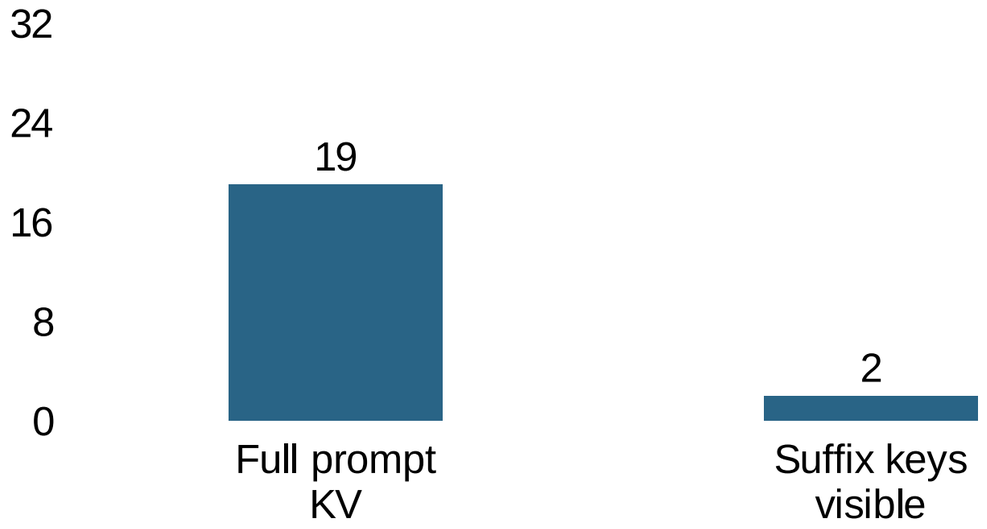
100 fixed offline helper calls. Full-KV success does not prove compact-memory sufficiency or successful ILM image generation.

Context visibility and native image continuation

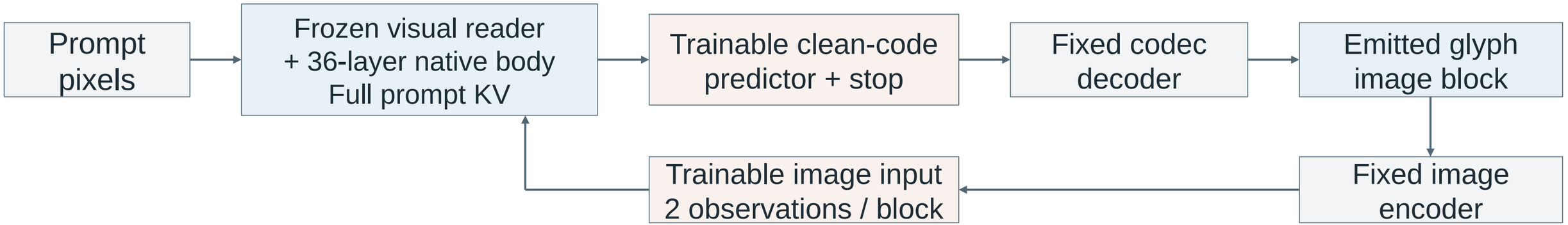
a Exact copy answers / 12 requests



b Exact answers / 32 training questions



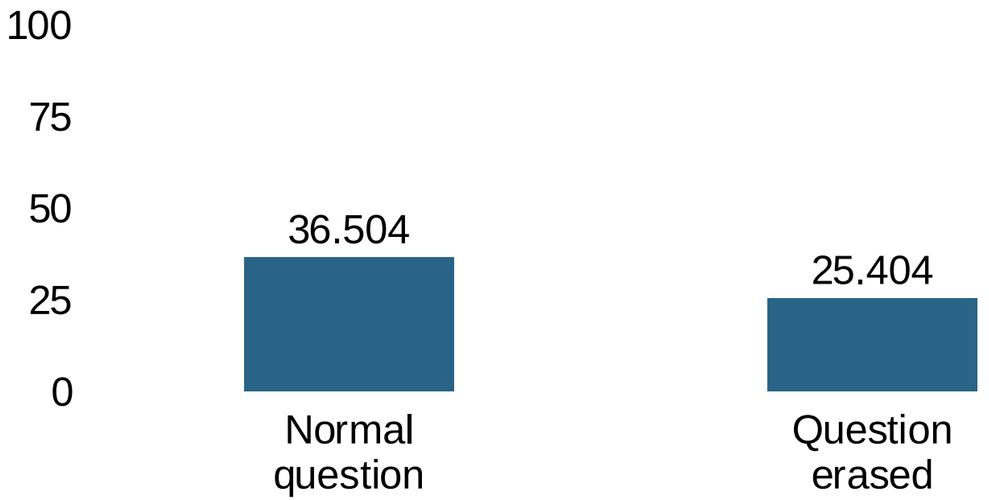
c New image actor: execution only; no trained language result



7.583B total parameters; 46.900M trainable interface parameters. External pretraining retained; 16.638 GiB execution peak.

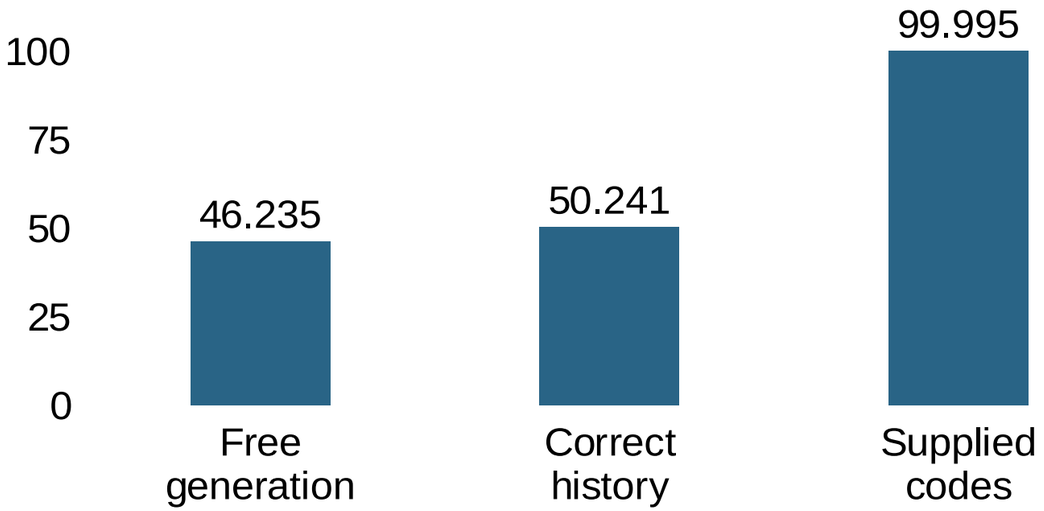
Conditional writing fails after full-context training

a Chinese development / F1 (%)



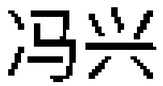
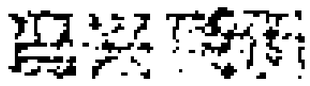
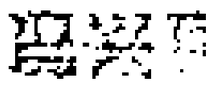
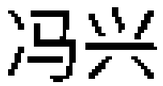
Exact images: 0/128 in both routes

b Training controls / F1 (%)



Exact images: 0/32, 0/32, 28/32

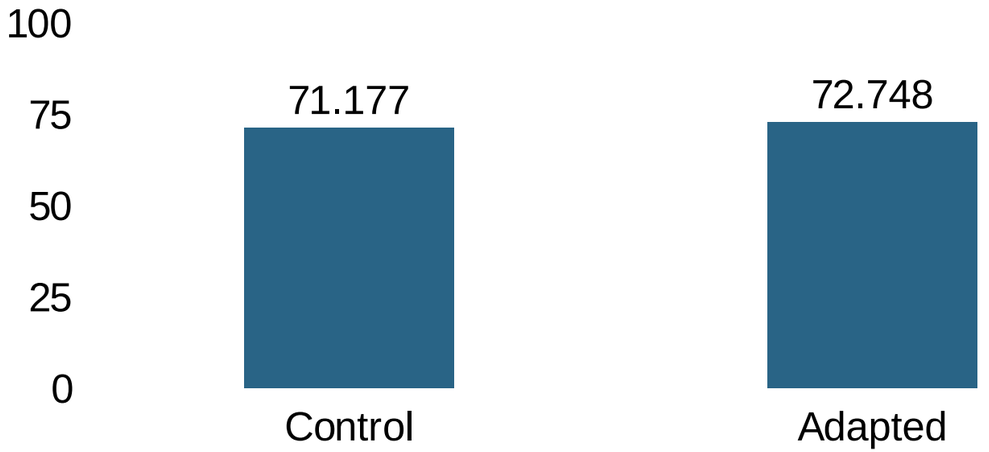
c First fixed question pair / complete raw writing images

	Reference	Free generation	Correct history	Supplied codes
Question A				
Question B				

40,000 updates; 7.583B total / 46.900M trainable parameters. Image agreement does not establish semantic accuracy.

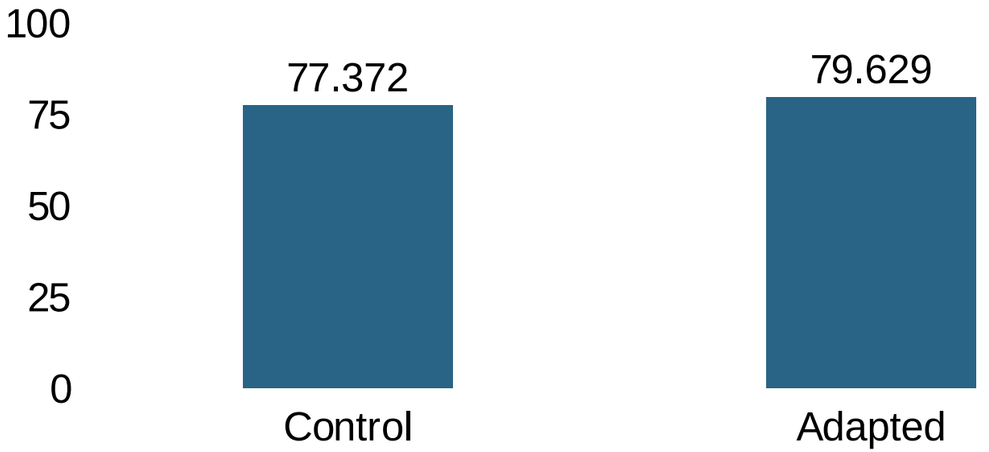
Trainable readout leaves autonomous writing inaccurate

a Autonomous writing / F1 (%)



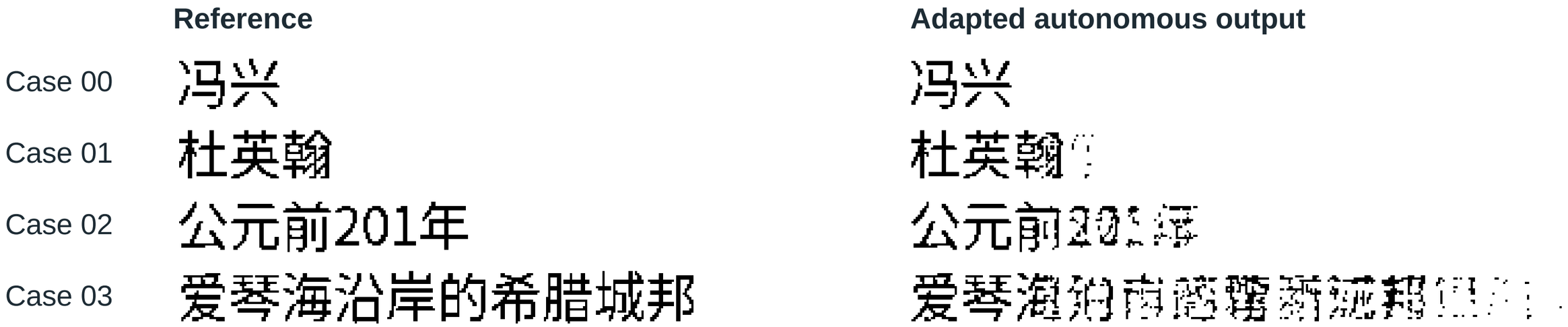
Exact complete images: 0/32 in both arms

b Correct preceding history / F1 (%)



Exact complete images: 0/32 in both arms

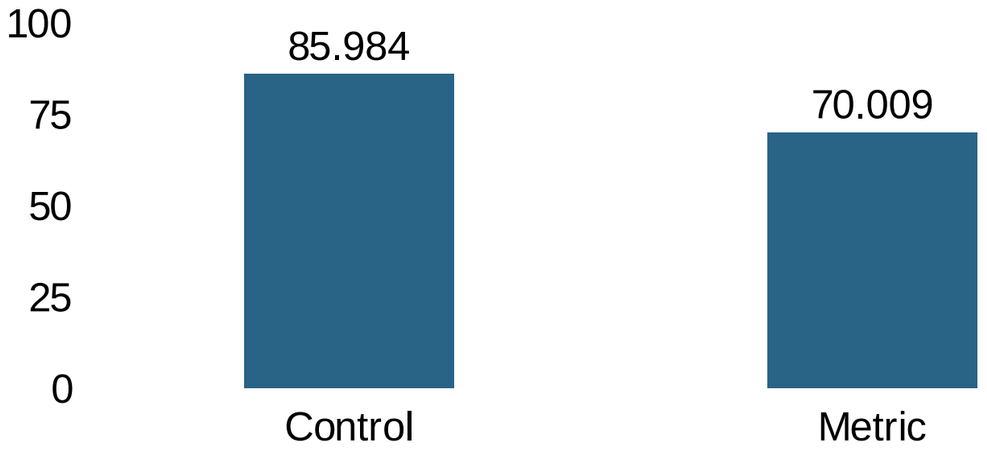
c First four fixed questions / complete raw images



32 training questions; 2,000 updates/arm. Supplied-code F1: 99.9953% (28/32 exact). Language remains unqualified.

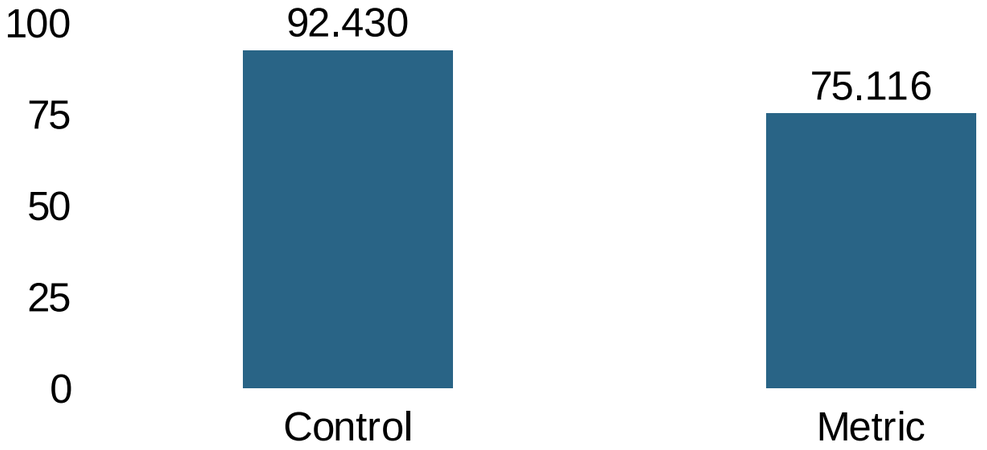
Decoder-sensitive loss reduces answer-image fidelity

a Autonomous writing / F1 (%)



Exact complete images: 0/32 in both arms

b Correct preceding history / F1 (%)



Exact complete images: 0/32 in both arms

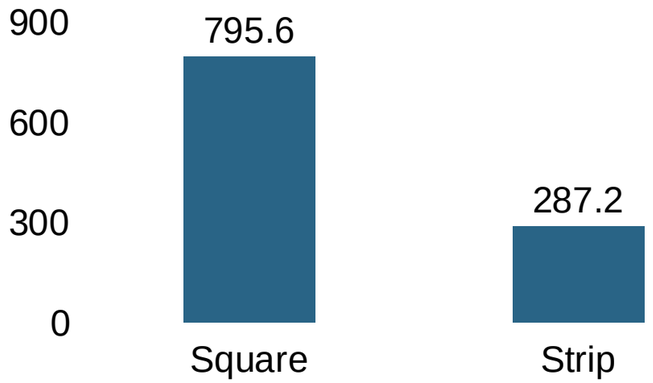
c First four fixed questions / complete raw images

	Reference	Control	Decoder metric
00	冯兴	冯兴	冯兴
01	杜英翰	杜英翰	杜英翰
02	公元前201年	公元前201年	公元前
03	爱琴海沿岸的希腊城邦	爱琴海沿岸的希腊城邦	爱琴海

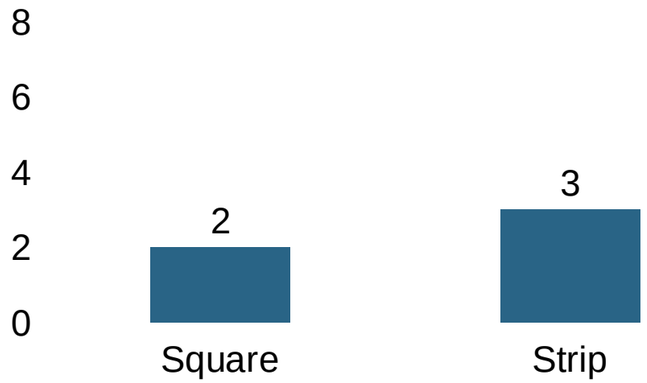
32 training questions; 2,000 updates/arm. All four repair gates fail. Supplied-code F1: 99.9953% (28/32 exact).

Paired image learning remains unreliable

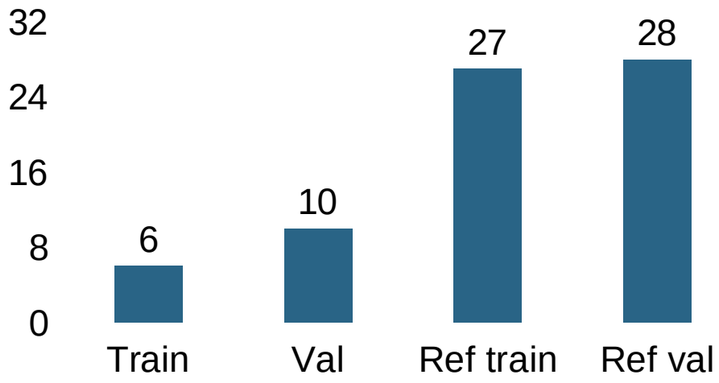
a Training invocation (s)



b Fixed screen: OCR exact / 8



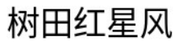
c Full strip: OCR exact / 32



d First four validation cases / full reference and generated canvases

Validation 00

Reference



Generated

Validation 01

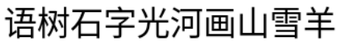
Reference



Generated

Validation 02

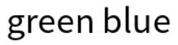
Reference



Generated

Validation 03

Reference

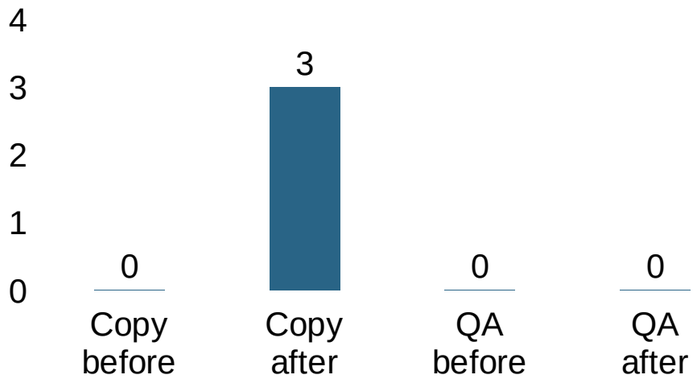


Generated

512 updates/arm on 32 copies; reference OCR: 27/32 train, 28/32 validation. No QA training or language-efficiency claim.

Visual reading after adapting the image frontend

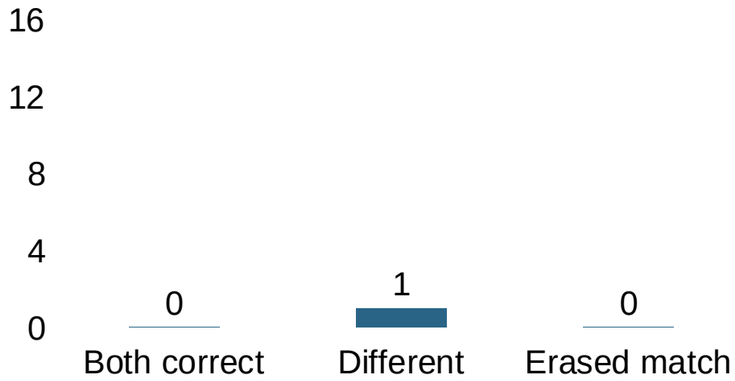
a Held-out screen: exact / 4



b Full validation: exact / 32



c Passage-pair controls / 16



d Source writing: supplied-string controls

花多地地山



mountain bright blue cloud



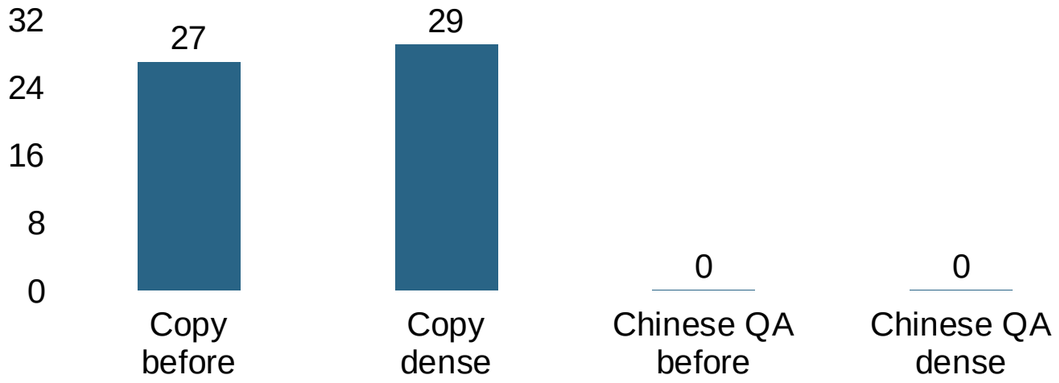
e First four held-out copies: auxiliary text readouts

Requested	After learning
树田红星风	树田红星风
garden silver open blue	garden silver open blue
语树石字光河画山雪羊	语树石字光河画山雪字
green blue	green blue

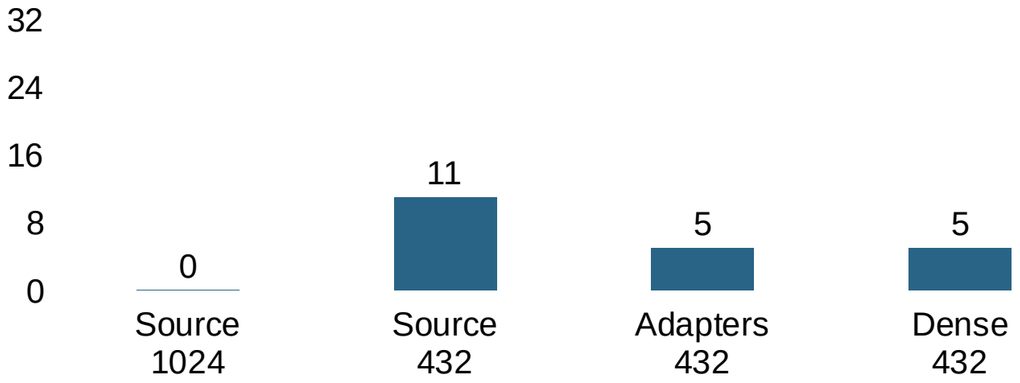
4,096 updates; 384 training examples; only visual adapters learned. Text readouts are not image answers. General ILM remains unachieved.

Native image geometry recovers source reading

a Original geometry: exact answers / 32



b English geometry check: exact / 32



c Same complete page at both input scales

1024 square: 4096 visual positions

Passage:
The programme's first serial, An Unearthly Child, shows that the Doctor has a granddaughter, Susan Foreman. In the 1967 serial, Tomb of the Cybermen, when Victoria Waterfield doubts the Doctor can remember his family because of, "being so ancient", the Doctor says that he can when he really wants to—"The rest of the time they sleep in my mind". The 2005 series reveals that the Ninth Doctor thought he was the last surviving Time Lord, and that his home planet had been destroyed; in "The Empty Child" (2005), Dr. Constantine states that, "Before the war even began, I was a father and a grandfather. Now I am neither." The Doctor remarks in response, "Yeah, I know the feeling." In "Smith and Jones" (2007), when asked if he had a brother, he replied, " No, not any more." In both "Fear Her" (2006) and "The Doctor's Daughter" (2008), he states that he had, in the past, been a father.
Question: What is the name of the first Doctor Who serial?
Answer:

432 square: 729 visual positions

Passage:
The programme's first serial, An Unearthly Child, shows that the Doctor has a granddaughter, Susan Foreman. In the 1967 serial, Tomb of the Cybermen, when Victoria Waterfield doubts the Doctor can remember his family because of, "being so ancient", the Doctor says that he can when he really wants to—"The rest of the time they sleep in my mind". The 2005 series reveals that the Ninth Doctor thought he was the last surviving Time Lord, and that his home planet had been destroyed; in "The Empty Child" (2005), Dr. Constantine states that, "Before the war even began, I was a father and a grandfather. Now I am neither." The Doctor remarks in response, "Yeah, I know the feeling." In "Smith and Jones" (2007), when asked if he had a brother, he replied, " No, not any more." In both "Fear Her" (2006) and "The Doctor's Daughter" (2008), he states that he had, in the past, been a father.
Question: What is the name of the first Doctor Who serial?
Answer:

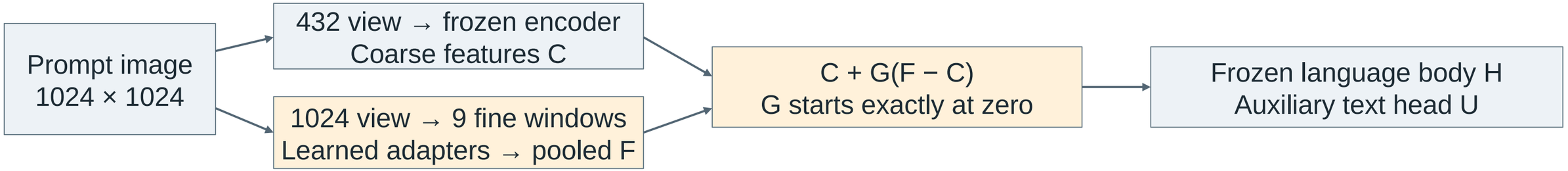
d First four source answers at 432: text readouts

Primary reference	Source answer
An Unearthly Child	The name of the first Doctor Who serial is An Unearthly Child.
Susan Foreman	SUSAN FOREMAN
7.5	7.5%
32	32%

Auxiliary Unicode readouts. Exact scores accept all annotated alternatives and require EOS. No trained image-answer result.

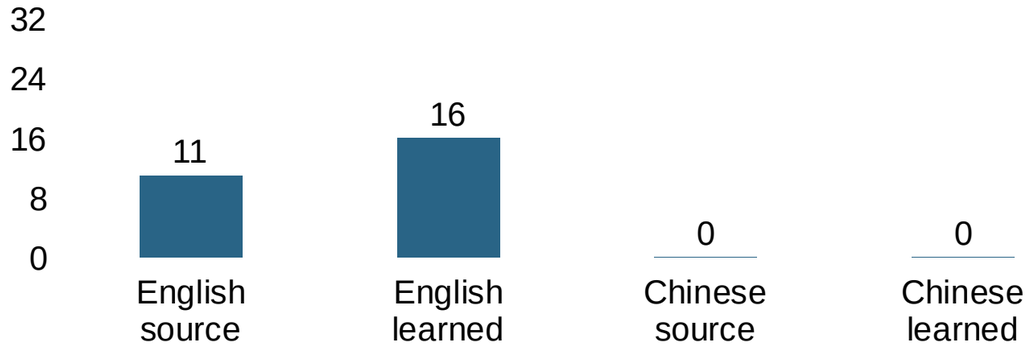
Visual-state learning and canvas dependence

a Learned correction to a frozen native reader



Training only: original-text pre-answer states at seven layers; answer CE + normalized state MSE

b Held-out QA: exact stopped answers / 32



Copying: English 7/16 → 16/16; Chinese 0/16 → 0/16

c Separate dense source: Chinese copies / 16



English copies: 16/16 → 7/16 → 0/16; QA stays 0/32

d Seen copying request; white display padding omitted

请只抄写引号内的内容：
“花多地地山”

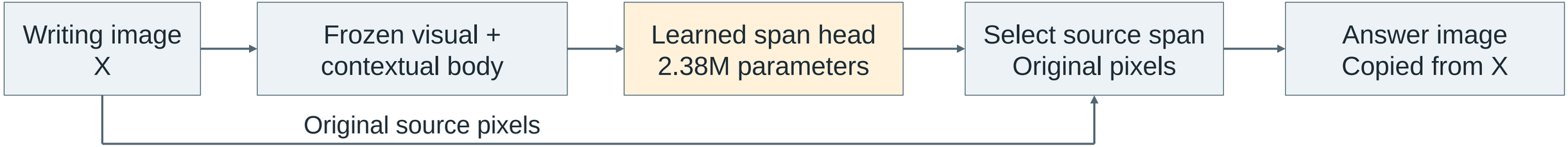
Raw auxiliary text readout (not an answer image)

Requested: 花多地地山
Learner: 花多梅树山
All 8 tested seen Chinese strings change characters

State correction: 4/6 requirements fail.
Attention-only: QA 0/32 both languages; 6/6 fail.

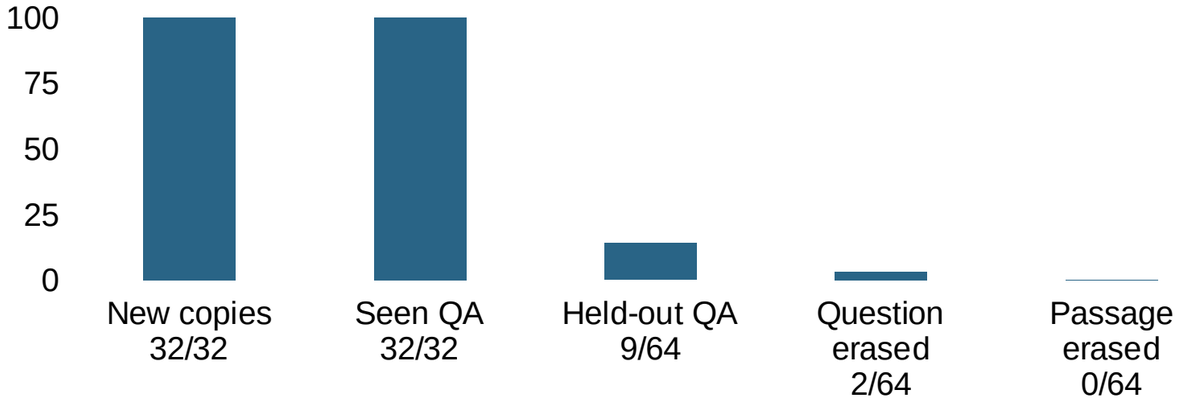
Exact copying with weak question answering

a Image-only contextual span selection



Normalized scores over visible spans; duplicate images share probability mass

b Exact image answers (%); counts below



Both questions correct: 1/32 paragraph pairs

3/4 requirements fail; source-copy component only

c First fixed held-out answer pair

1. What are the fish's common names?

Reference 1

国公、白虎

Output 1: wrong

为辐鳍鱼纲鲀形目鲀亚目，旧属鲀科

2. Which ocean regions does it inhabit?

Reference 2

印度洋及西太平洋区

Output 2: correct

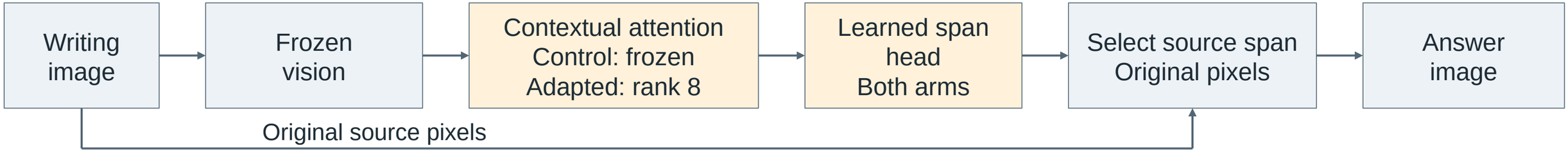
印度洋及西太平洋区

7.531B full runtime with external pretraining; 320 training images, 3,200 updates

Source copies cannot write answers absent from the input; free generation remains unachieved

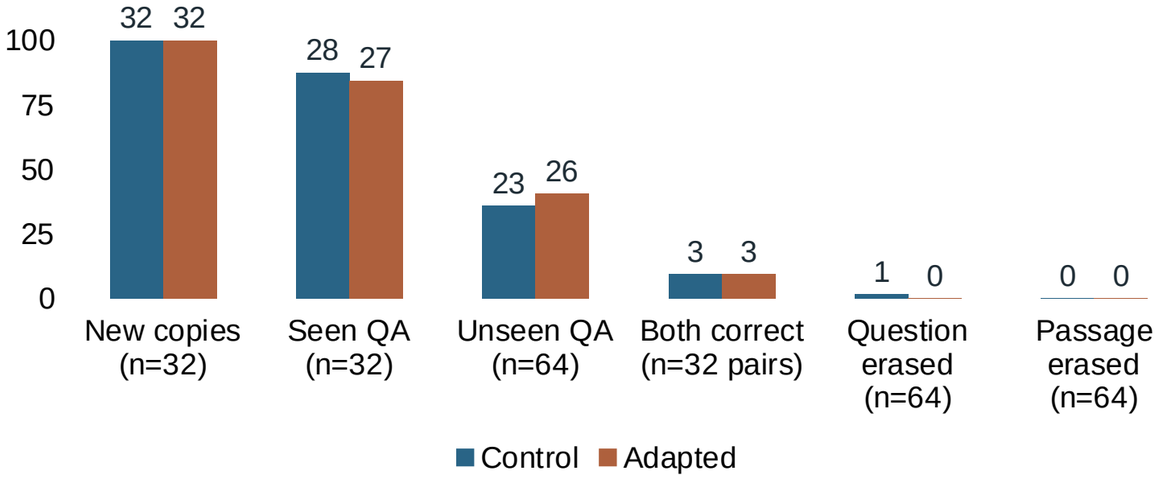
Contextual learning improves few answers and misses reader requirements

a Same source-pixel motor; matched learning schedule



3,257 updates per arm; 2.378M / 8.572M learned parameters; 7.531B pretrained full runtime

b Exact image answers (%); labels give correct counts



Gain: +3 answers, +0 complete pairs

Both fail reader floors; adaptation misses improvement rule

c First fixed unseen question pair

1. What are the fish's common names?

Reference	国公、白虎
Control	国公、白虎，
Adapted	国公、白虎，

2. Which ocean regions does it inhabit?

Reference	印度洋及西太平洋区
Control	印度洋及西太平洋区
Adapted	印度洋及西太平洋区

Both add a comma in Q1; both reproduce Q2 exactly

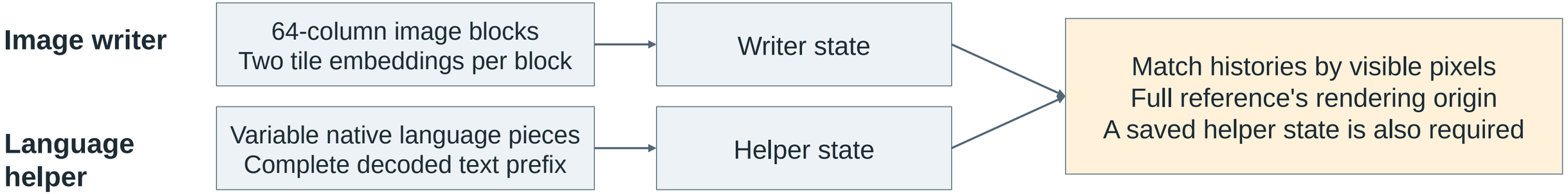
d Separate task-memory substitution: earlier head

53/64 development selections change; original score 9/64. Hybrid features, not native new prompts.

Source crops cannot compose absent answers. Earlier-head substitution is not a third comparison arm.

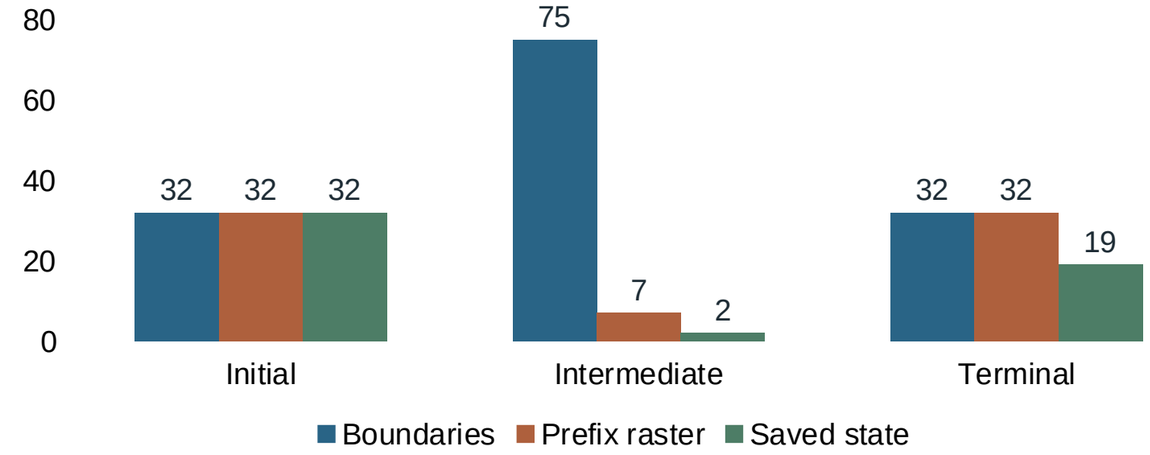
Image steps and language pieces usually describe different histories

a Audit identical preceding content before comparing states



Equal ordinal step does not establish equal history

b All retained boundary counts



All 32 initial states agree after common BF16 rounding

c Both eligible intermediate matches

TRAIN_1779_QUERY_1

美国维吉尼亚州的

Image positions: 6 Helper positions: 6

Normalized state distance: 1.497 (uncalibrated)

TRAIN_2146_QUERY_1

1.7-2.

Image positions: 2 Helper positions: 6

Normalized state distance: 1.407 (uncalibrated)

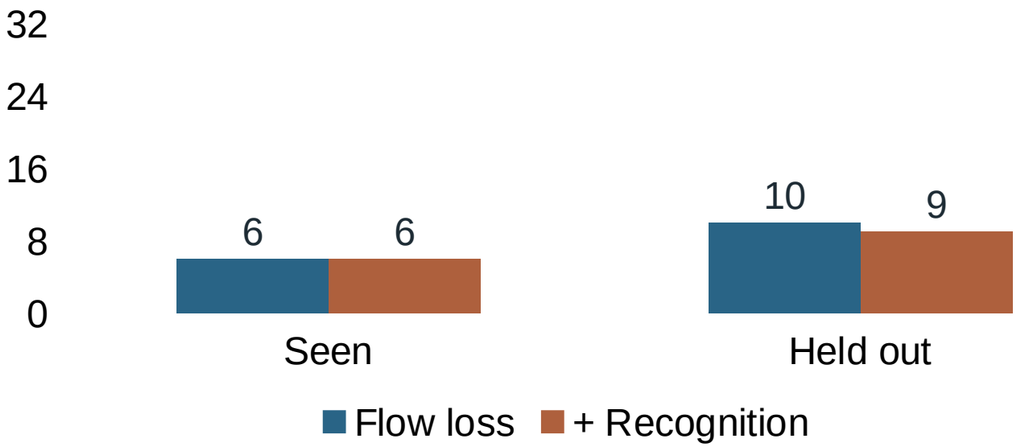
68/75 intermediate boundaries have no complete-prefix raster; 5 more lack a saved matching state

All 216 saved source-prefix positions decode to complete UTF-8; 13 terminal states are missing

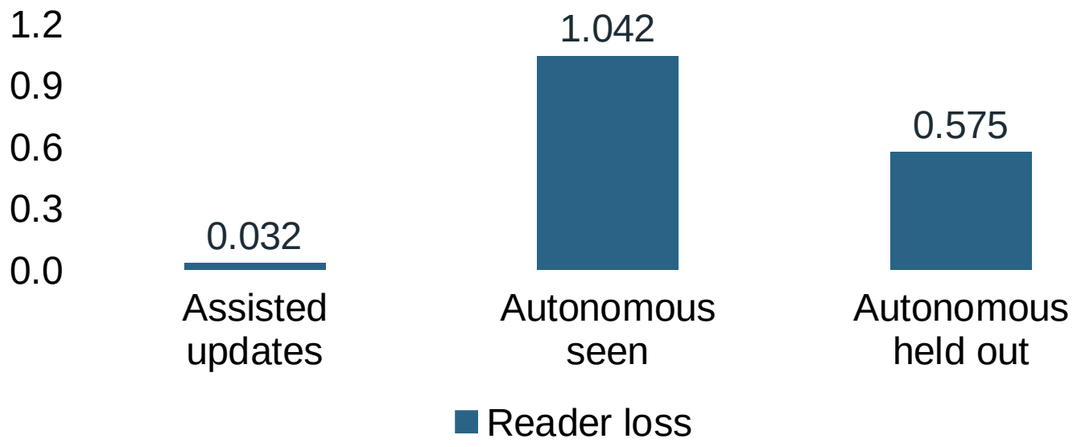
Existing traces only: zero new neural calls, zero updates. No alignment treatment or writer repair tested.

Recognition loss does not ensure faithful visible writing

a Independent OCR-exact copies (32 per split)



b Training-reader target NLL (nats)



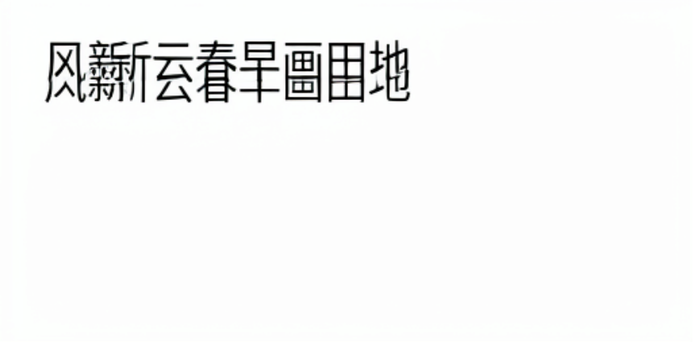
255 assisted updates and final-state free outputs are different regimes; this is not a causal decomposition.

c Reader ignores a visible duplicated line

Reference

风新云春早画田地

Actual generated page



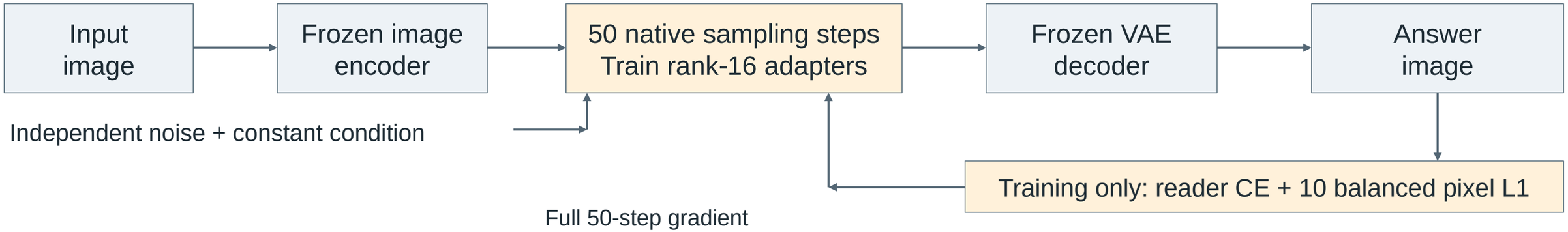
Reader returns the expected string.
Gold NLL: 0.0424; clean reference: 0.000192.
Recognition does not count every visible repetition.

6/32 and 9/32 with the independent evaluator; 14/32 and 19/32 with the training reader.
The reader is not an independent judge. No reliable writer or semantic QA is established.

Same first 32 image pairs; 512 updates; source backbone and condition fixed. Complete 64-case outputs retained.

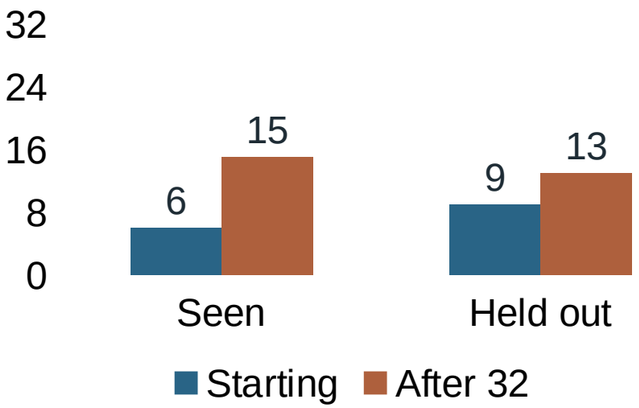
Autonomous image-to-image learning remains below reliable writing

a Full sampling trajectory; targets enter only the final loss



32 updates: 1,135.452 s; 12.057 GiB peak allocated. Runtime: 3.900B transformer + 84.046M VAE.

b OCR-exact copies (n=32)



Progress rule met; 30/32 writing floor not met.
No matched extra-training control.

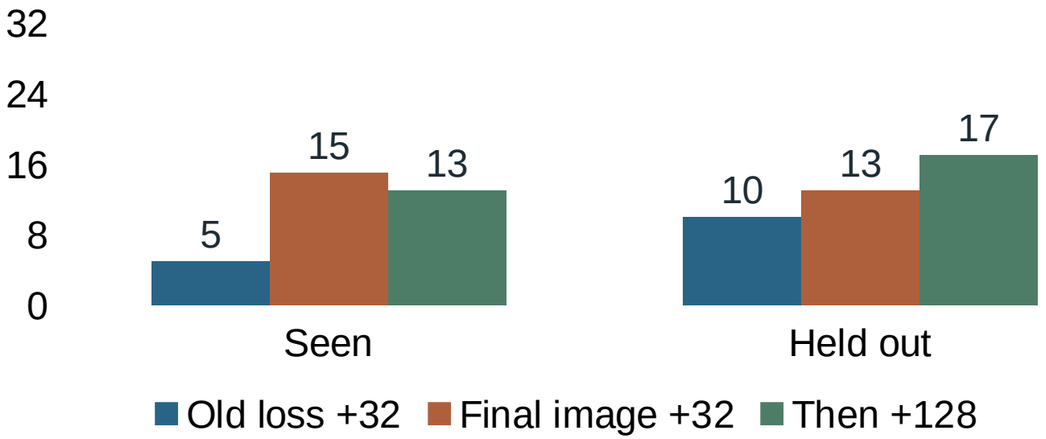
c First two held-out inputs: actual pages

Target	Starting	After pilot
树田红星风	树田丝红星风”	树田红工星风”
garden silver open blue	garden silvver ope blue	garden silver open blue

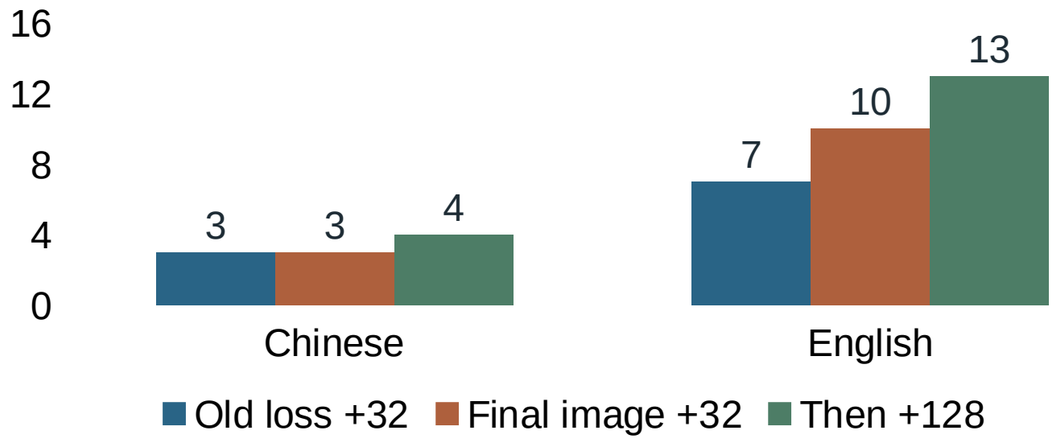
Unwanted punctuation remains in Chinese; the English phrase restores its missing letter.

Balanced exposure and autonomous image writing

a Complete OCR-exact copies (32 per split)



b Held-out copying by language (16 each)



128 additional updates
Four complete passes over 32 examples
64 Chinese and 64 English draws

Training: 4,500.1 s
Peak: 12.032 GiB
Full 50-step trajectory gradient

Continuation rule: not met
Copying does not establish semantic QA.
Extra updates and exposure change together.

c First two held-out inputs: complete actual pages

Target	Starting pilot	After +128
树田红星风	树田红工星风”	树田红工星风”
garden silver open blue	garden silver open blue	garden silver open blue

Canonical writing style; reused development cases; inherited 3.900B transformer + 84.046M VAE.